

GENERAL SPECIFICATION G3.13

Bearing Piles and Sheet Piling

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G3.13 BEARING PILES AND SHEET PILING

G3.13.1 Scope

- (a) This Section contains requirements which, where relevant to this Contract, shall apply to bearing piles of the following types:
- (i) Precast reinforced concrete;
 - (ii) Precast prestressed concrete;
 - (iii) Driven cast in place concrete;
 - (iv) Bored cast in place concrete;
 - (v) Continuous flight augured cast-in-place concrete;
 - (vi) Steel piles of H section or box section;
 - (vii) Pressure impregnated timber; and
 - (viii) Bakau;
- (b) This Section also contains requirements which, where relevant to this Contract, shall apply to steel sheet piling for use in the Permanent Works, including any permanent or temporary bracing, ties and other ancillary works needed for the steel sheet piling.

G3.13.2 Definitions

- (a) The following terms shall have the meanings hereby assigned to them:
- | | |
|---------------------|--|
| "Bearing piles" | Piles supporting structures in the Permanent Works. They may be required to resist vertical loads in a downward or upward direction, or both. Some bearing piles may be raked to resist lateral loads. |
| "Steel sheet piles" | Fully interlocking steel piles that can be driven to form a continuous wall or diaphragm. |
| "Bracing" | Includes walings, struts and connections of all types; and |
| "Ties" | Includes their anchorages and connectors. |

G3.13.3 Reference Standards

- (a) Unless otherwise specified, bearing piles shall comply with the relevant Singapore Standards and the Reference Standards listed below, where applicable:

Standard	Subject
SS CP4	Foundations
SS EN 1997	Eurocode 7: Geotechnical Design
BS 639	Covered carbon and carbon manganese steel electrodes for manual metal- arc welding
BS 4449	Carbon steel bars for the reinforcement of concrete
BS 4486	Hot rolled and processed high tensile alloy steel bars for the pre-stressing of concrete

Standard	Subject
BS 4871	Approval testing of welders working to approved welding procedures
BS 5896	High tensile steel wire and strand for the pre-stressing of concrete
BS 6949	Bitumen based coatings for cold application excluding use in contact with potable water
BS 812	Testing aggregates
BS 8102	Protection of structures against water from the ground
SS EN 206-1	Concrete: Specification, performance, production and conformity
SS 544	Complimentary Singapore Standard to SS EN 206-1
BS 8666	Scheduling, dimensioning, bending, and cutting of steel reinforcement for concrete
BS EN 1011	Welding. Recommendations for welding of metallic materials
SS EN 1990	Basis of structural design
SS EN 1991-1-5	Actions on structures: Thermal actions
SS EN 1991-1-6	Actions on structures: Actions during execution
SS EN 1992	Design of concrete structures
SS 197-1	Cement: Composition, specification, and conformity criteria for common cements
BS EN 12350	Testing fresh concrete
BS EN 12390	Testing hardened concrete
BS EN 10080	Steel for reinforcement of concrete
SS 560	Steel for reinforcement of concrete – Weldable reinforcing steel – Bar, coil and decoiled product
SS 561	Steel fabric for the reinforcement of concrete
BS EN ISO 17660	Welding – Welding of reinforcing steel
BS EN 13670	Execution of Concrete Structures
BS EN 13791	Testing concrete
BS EN ISO 15630	Steel for the reinforcement and pre-stressing of concrete: test methods
BS EN 15743	Supersulphated Cement
BS 6213	Guide to selection of construction sealants

G3.13.4 Submissions by the Contractor

- (a) Programme
 - (i) The Contractor shall submit for approval to the S.O. his proposed programme for the execution of the piling work at least seven (7) days before commencement of the work. In addition, the Contractor shall inform the S.O. daily of the programme of piling for the following working day and shall give adequate notice of his intention to work outside normal hours, if this has already been approved by the S.O..
- (b) Piling Equipment
 - (i) Before any piling work is commenced, the Contractor shall submit for approval to the S.O. full details of the pile driving equipment and the method of carrying out the work which he intends to use. Such information shall include a full description of the piling frame, hammer, helmet and packing and of a method of handling and pitching piles and supporting them during driving, and of the proposed driving procedure to give penetration to the required level and of the proposed set for the working load on the pile and the method of calculating it. Any revisions to these proposals which, in the light of ensuing experience, appear desirable, shall also be submitted for the approval of the S.O. prior to implementation of said revisions. The Contractor shall submit all of his proposals at least three weeks before the date on which he intends to use the plant on the Site.
- (c) Submissions for Bearing Piles
 - (i) Submissions required from the Contractor in relation to the design, installation and testing of bearing piles include the following:
 - 1) Within four weeks of Acceptance of Tender for the approval of the S.O.
 - Detailed design calculations;
 - Name of proposed supplier, or proposed methods of manufacture, as relevant; and
 - Proposed methods of installation.
 - 2) At time of Installation
 - Detailed records of installation as carried out, which, as a minimum, shall include: location, top of pile level, toe level, and variation between design and actual parameters.
 - 3) Within one week of Completion of Tests
 - Reports of test results.
- (d) Submissions for Steel Sheet Piling
 - (i) Submissions which the Contractor is required to make in relation to steel sheet piling include, where relevant, the following:
 - 1) Steel sheet pile section proposed;
 - 2) Proposals for storing, handling and transporting steel sheet piles;
 - 3) Records of piling;

- 4) Detailed design calculations;
 - 5) Steel grade;
 - 6) Proposed supplier; and
 - 7) Relevant certification.
- (e) As-Constructed Records and Drawings
- (i) After the completion of the piling works and on the S.O.'s instruction, the Contractor shall within a reasonable period produce and hand over to the S.O. 'As-Constructed' piling Drawings showing all necessary and desirable details of all those piled foundations as decided by the S.O. and installed on Site by the Contractor.
 - (ii) These 'As-Constructed' piling Drawings shall be in microstation form and shall show the 'As-Constructed' sizes, types, cut-off levels, penetration, positions, inclinations, dimensions, eccentricities, final setting and numbers of all piles, pile caps, all reinforcement and any other information the S.O. require for record purposes. One (1) hard copy of A1 size shall be submitted to the S.O..
 - (iii) These 'As-Constructed' Drawings showing the actual sizes, type, levels, penetration, positions, inclinations and eccentricities of the piles on both directions and their cut-off levels shall be duly endorsed by a Licensed Surveyor.
 - (iv) All 'As-Constructed' Drawings shall be duly endorsed by a Professional Engineer to confirm their accuracy and that all details are constructed in accordance with approved Drawings.
 - (v) The contractor shall submit to S.O. the complete piling record book (2 sets) properly bound and titled.
- (f) The results of the soil investigation are solely for the purpose of assisting the Contractor in the preparation of his tender. Neither the Board nor the S.O. bear any responsibility for the correctness of the result.
- (g) It shall be the Contractor's responsibility to carry out further sub-surface investigation if he has any doubt about the available information or if he considers further information necessary or desirable. Such investigation by the Contractor shall be entirely at his own expense and shall have the S.O.'s prior approval in writing. The S.O. shall be supplied free of charge with two (2) copies of the results of all such investigation by the Contractor.
- (h) In addition to submitting the record required, the Contractor shall report immediately to the S.O. any circumstances which indicate that the ground conditions differ from those expected by the Contractor from his interpretation of the results of soil investigation carried out previously and which affect adversely the bearing capacity of the pile.
- (i) The Contractor's attention is, however, drawn to the fact that suitable founding strata may be encountered at depths which require the actual installation of different lengths of pile and which may give rise to considerable variation within any particular site or pile groups.

- (j) In the event of unforeseen obstructions below ground level being encountered during boring, the Contractor shall immediately inform the S.O.. If in the opinion of the S.O. further progress on that pile is impractical by the method employed, the S.O. shall instruct the Contractor for the necessary action to be taken. Where the S.O. instructs the Contractor to abandon the pile from this cause and proceed with piling work on another pile, the Contractor will be paid only for this length of empty boring and backfilling.

G3.13.5 Piling Equipment

- (a) The piling equipment shall be capable of speedily and efficiently installing piles of the specified types, sizes, lengths and safe working loads.
- (b) The type of piling equipment shall be suitable for the type of piles to be driven and shall be that of a system which produces the minimum noise during driving.
- (c) The Contractor shall provide sufficient number of units to ensure completion of piling work according to programme.
- (d) The piling frame shall be of sturdy construction, supported on an adjustable base, securely guyed and with ample toggle connections to leaders so that the pile is firmly held at all times. The type and weight of hammer shall be to the approval of the S.O. and the weight of the hammer shall be at least half that of the pile or otherwise specified. In general, a heavy hammer with a short drop should be used in preference to a light hammer with a longer drop. Single or double acting diesel or air operated hammers may also be employed. The Contractor shall submit confirmation of hammer energy per blow to the satisfaction of the S.O.. All plant being used shall be maintained in a satisfactory condition and any items suffering wastage or damage shall be promptly replaced or repaired.
- (e) No piling shall be carried out without the approval in writing of the S.O. of the equipment and method of working and any revisions to these as described above.

G3.13.6 Pile Setting Out

- (a) Contractor to install marker pins at the pile positions by a registered land surveyor and these positions shall be verified before installation of piles. However, the acceptance of the final position of the piles shall not relieve the Contractor of his responsibility of all necessary remedial works ordered by the S.O. as a result of inaccuracy in the position of piles installed.

G3.13.7 Pile Driving

- (a) Unless otherwise specified, pile driving shall be as specified hereunder.
- (b) The driving equipment to be used shall be of such type and capacity to the approval of the S.O.. If drop hammer is used, it shall be of a free fall type and the weight of the hammer shall be as specified in the relevant standard.
- (c) Each pile for driving shall be marked clearly in paint with its number and its overall length. In addition, each pile shall be marked at intervals of 500mm to enable easy determination of pile penetration. Where necessary, the intervals shall be reduced to 250mm as directed by the S.O..

- (d) Piles shall be pitched accurately in the positions as shown on the Drawings. At all stages during driving and until the pile has been or driven to the required depth or set, all exposed piles shall be adequately supported and restrained by means of leaders, trestles, temporary support or other guide arrangements to maintain position and alignment and to prevent buckling and damage to the piles.
- (e) Each pile shall be driven continuously until the specified set and/or depth has been reached. However, the S.O. may permit the suspension of driving if he is satisfied that the rate of penetration prior to the cessation of driving will be substantially re-established on its resumption or if he is satisfied that the suspension of driving is beyond the control of the Contractor.
- (f) A follower (long dolly) shall not be used for driving end bearing piles. It may be used for driving frictional piles with prior approval of the S.O..
- (g) The Contractor shall inform the S.O. without delay if an unexpected change in driving characteristics is encountered.
- (h) Where required by the S.O. set shall be taken at approved intervals during the driving to establish the behaviour of the piles. A set shall be taken only in the presence of the S.O. unless otherwise approved. The Contractor shall provide all facilities to enable the S.O. to check driving resistances.
- (i) Re-drive checks, if required, shall be carried out in accordance with a procedure approved by the S.O..
- (j) A detailed record of the driving resistance over the full length of each pile shall be kept. The log shall record the number of blows for every 300 mm of pile penetration.
- (k) When a final set is being measured, the following requirements shall be met:
 - (i) The exposed part of the pile shall be in good condition, without damage or distortion;
 - (ii) The dolly packing shall be in sound condition;
 - (iii) The hammer blow shall be in line with the pile axis and the impact surfaces shall be flat and at right angle to the pile and hammer axis;
 - (iv) The hammer shall be in good condition and operating correctly; and
 - (v) The temporary compression of the pile shall be recorded if required.
- (l) Piles shall be driven in an approved sequence to minimise the detrimental effects of heave and lateral displacement of the ground. When required, levels and measurements shall be taken to determine the movement of the ground or any pile resulting from the driving process. If any pile rise occurs as a result of adjacent piles being driven, the Contractor shall submit to the S.O. his proposals for correcting this and to avoid the same in subsequent work. The cost associated shall be deemed inclusive in the contract sum.

G3.13.8 Bearing Piles

- (a) Design Responsibility
 - (i) The Contractor shall install the bearing piles at the design positions and to the rakes as shown on the Drawings. Where the type of piles is not shown on the Drawings, the Contractor may propose any of the types listed above for the approval of the S.O..
- (b) Design and Materials
 - (i) General - Bearing Piles of all Types
 - 1) Bearing piles shall be designed to carry safely the working loads shown on the Drawings. The ultimate bearing capacity, as determined by test in accordance with the general specifications, shall be at least twice the working load as defined by the S.O.. The ultimate bearing capacity is defined as the load which, applied at the head of the pile, causes the head of the pile to settle a distance equal to one-tenth of the pile diameter. In the case of non-circular piles, 'diameter' shall be taken to mean the diameter of the circle having an area equal to the cross-sectional area of the pile or, in the case of steel H-piles, to the area bounded by the flanges and lines joining the edges of the flanges.
 - 2) The approximate length of bearing piles shall be determined from any geological information included in the Specification or obtained by preliminary site investigation or shall be determined from load tests carried out on non-working piles.
 - 3) The final length of driven piles shall be determined by driving to a 'final set', expressed as the penetration in mm per 10 blows, approved by the S.O.. The final set may be calculated using an established pile-driving formula but shall not exceed the average of final sets recorded when driving non-working piles which have achieved the required ultimate bearing capacity.
 - 4) The final length of bored or augured piles shall be not less than the average length of non-working piles which have achieved the required ultimate bearing capacity.
 - 5) Precast concrete piles and pile casings shall be designed to develop the strength necessary to withstand without damage the stresses arising during transportation, handling and driving. The design shall be such that the compressive stress in the concrete during handling and pitching and under the working load does not exceed $0.275 u$ due to direct load or $0.366 u$ due to bending, where u is the 28-day works cube stress.

(ii) Precast Reinforced Concrete Piles

- 1) Depending upon the nature of the ground through which the piles are to be driven, and the nature and concentrations of dissolved salts in the groundwater, concrete for use in piles shall be made with ordinary Portland cement, sulphate-resisting Portland cement or super-sulphated cement, following the guidelines given in SS EN 1992-1. The maximum size of aggregate shall be 20 mm and the minimum cement content 370 kg/m³ of finished concrete. Except where otherwise specified, the maximum free water/cement ratio shall be 0.4. The minimum cube strength at 28 days shall be 45 N/mm² and the maximum cube strength at 28 days shall be 60 N/mm².

Concrete mixes shall be specified in accordance with the requirements of the Specification and the relevant Eurocode Standards.

- 2) The cover to all reinforcement including binding wire shall be not less than 50 mm. Hoops and links shall fit tightly against longitudinal bars and be bound to them by welding or soft iron wire with the free ends turned inwards. The longitudinal bars shall be held apart by spreader forks not more than 1.5 m apart.
- 3) The main longitudinal bars shall be level at the top of the pile and fit tightly into the shoe if one is used. Joints in the main longitudinal bars shall be made by full penetration butt welding in accordance with BS EN 1011-2:2001.
- 4) Piles shall be provided with flat or pointed co-axial shoes of steel or cast iron if the concrete at the tip of the pile is liable to be damaged by driving.
- 5) Reinforcement shall comprise mild steel deformed bars to BS 4449 and shall meet the following minimum requirements:
 - Longitudinal reinforcement shall be at least 1.0% of the gross concrete area;
 - Lateral reinforcement shall be in the form of hoops or links having a diameter of at least 6 mm; and
 - The volume of lateral reinforcement shall be at least 0.2% of the gross volume except near the top of the pile where, for a distance of three times the width/diameter of the pile measured from the top of the pile, the volume of lateral reinforcement shall be not less than 0.6% of the gross volume, and for an equal distance below this where there shall be a gradual transition in the lateral reinforcement from at least 0.6% of the gross volume to at least 0.2%.
- 6) Manufacture, curing, transport and storage shall be in accordance with the recommendations of the relevant standard.

(iii) Precast Prestressed Concrete Piles

- 1) The cover to prestressing steel and reinforcement shall be not less than 50 mm.
- 2) Concrete shall be as specified for precast reinforced concrete piles above.
- 3) Driven Cast in-place Piles.

- 4) Driven cast in place piles shall have a casing of steel or concrete which shall be driven to a set and filled with concrete. The casing may, and where specified shall, be left in place as part of the Permanent Works. The casing shall be suitable for the proposed method of installation and for the purpose of forming a pile.
- 5) Precast concrete casing shall be reinforced either with steel bar or mesh or with polypropylene fibres. Concrete casings and shoes shall be marked with a reference number and the date at the time they are made. They shall not be used in the Works until they are at least 28 days old.
- 6) Concrete casing may be delivered to the Site in short lengths ready for jointing. The use of damaged or cracked lengths of concrete casing will not be permitted.
- 7) Steel casing shall be of mild steel and shall be delivered to the Site in as long lengths as the Contractor can reasonably handle.
- 8) Concrete for filling the casing maximum free water/cement ratio shall be 0.45. It shall have adequate workability for the method of placing used to form the pile. The average compressive stress under working load shall not exceed 25% of the specified works cube strength at 28 days calculated on the total cross sectional area of the pile.
- 9) Unless otherwise specified, the cast-in-place concrete shall be reinforced over the whole length with mild steel deformed bars to SS EN 1992-1. The Contractor shall provide such steel as is necessary to comply with the provisions of BS 8110 for resisting stresses with a minimum longitudinal reinforcement of 0.8% of the gross concrete area in the top 3 m of the pile. The cover to all reinforcement, including binding wire, shall be not less than 50 mm measured to the inside of the casing, suitable spacers being provided to ensure this. Joints in the main longitudinal bars shall be made by full penetration butt welding.

(iv) Bored Cast in-place Piles

- 1) Bored cast in place piles shall have a casing of steel which shall be filled with concrete. The casing may, and where specified shall, be left in place as part of the Permanent Works. The casing shall be suitable for the method of installation and for the purpose of forming a pile.
- 2) Unless otherwise specified, piles shall be reinforced over the whole of their length.

(v) Continuous Flight Augured Piles

- 1) Continuous flight augured piles are piles formed by boring with a continuous flight auger and pumping concrete or grout down the hollow stem of the auger during retraction.
- 2) Concrete maximum free water/cement ratio may be increased as necessary to achieve the required workability. The mix shall be so designed that there is no segregation and only a minimum of bleeding.

- 3) Unless otherwise specified, the upper part of the piles shall be reinforced down to the depth specified or approved. Reinforcement shall be made up into cages designed to be inserted through the already-placed concrete or grout.

(vi) Steel Bearing Piles

- 1) Steel bearing piles shall be either of H beam section or fabricated box piles.
- 2) The cross sectional area of the steel shall be sufficient to withstand the blows of the hammer during driving and to carry the working load on the pile. The stress in the steel at working load shall not exceed 30% of the yield stress. Where the finished pile projects above the ground level the working stresses shall be reduced by the appropriate buckling coefficient.
- 3) The working load shall be transmitted to the pile through a capping plate, dowel bars or cleats.
- 4) Unless otherwise specified, fabricated box piles shall be filled with concrete, made with sulphate resisting cement and with aggregate of 20 mm maximum size. The minimum cement content shall be 330 kg/m³ and the maximum free water/cement ratio shall be 0.50. Minimum works cube strength at 28 days shall be 25 N/mm².
- 5) The bottom end of box piles shall be closed with a welded flat plate or a shaped shoe of cast iron, cast steel or fabricated steel.

(vii) Pressure Impregnated Timber Piles

- 1) All square sawn timber piles shall be Kempas or Keruing, of approved quality, strength and cross section. The piles shall be free from defects which may affect their strength and durability. All timber piles shall be inspected before driving to ensure compliance with these requirements.
- 2) The timber used shall conform to the following grading limits and specification:
 - Knots:
 - o Measurement: The size of a knot shall be taken as the maximum diameter of its visible cross section, no matter where it occurs. Where the knot can be seen in longitudinal section as well as in cross section, the longitudinal section shall be ignored. Where two or more knots occur in the same cross section, each knot shall be measured as before described and the sum of these measurements taken as the diameter of an equivalent single knot. Where only a portion of the cross section is visible, the maximum dimension of that portion shall be measured.
 - o Grading Limits: The size of knots shall not exceed one quarter the width of the face or edge on which they appear, nor shall knots be grouped closer than one knot per 1 m of the length of the pile. Unsound knots shall not be permitted.

- Wane:
 - o Measurement

The extent of wane is determined by adding together the widths of wane on any two adjacent faces.
 - o Grading Limits

Wane shall be allowed on one edge only and the total width of wane on its adjacent faces shall not exceed:

50 mm for 100 x 100 mm piles
65 mm for 125 x 125 mm piles
75 mm for 150 x 150 mm piles
90 mm for 175 x 175 mm piles

3) Curvature:

- Measurement: The extent of curvature is determined by measuring the maximum deviation from a tightly stretched string or wire joining the mid-points of butt and toe.
- Grading Limits: In piles, up to 6 m in length, maximum deviation shall not exceed 25 mm i.e. the wire/string shall fall within 25 mm of the centre line of the pile. In piles over 6 m long, maximum deviation shall not exceed 35 mm.

4) Borer Holes and Pin Holes:

- Measurement: The size of a borer hole shall be taken as the maximum diameter of its visible cross section, no matter where it occurs. Where only a portion of the cross section is visible, the maximum dimension of that portion shall be measured. Note: The area for measurement shall be determined from the full width of the piece and the length.
- Grading Limits: A pile shall not have holes which exceed the amount given in the following:
 - o Pin holes (up to 1.6 mm diameter) 32 Nos. over an area of 0.01m²;
 - o Shot holes (1.6 mm - 3 mm diameter) 4 Nos. over an area of 0.10m²; and
 - o Borer holes (3 mm - 6 mm diameter) 2 Nos, over an area of 0.10m².

5) Open Shakes and Surface Checks:

- Measurement: The size of an open shake or surface check shall be taken as its depth projected onto a surface at right angles to the face on which it occurs.
- Grading Limits: A pile shall not have shakes or checks whose depth expressed as a fraction of the width of the face at right angles to the one on which the defects occur which exceed 1/4.

- 6) End Splits:
 - Measurement: The size of an end split shall be taken as its length.
 - Grading Limits: The longest split shall not exceed 75 mm at each end.
- 7) Compression Failure:
 - Compression failure shall not be accepted.
- 8) Brittle Heart and Boxed Heart:
 - Brittle heart and boxed heart shall not be permitted.
- 9) Sloping Grain:
 - Measurement: The slope of grain shall be measured over a distance sufficiently great to determine the general slope, disregarding slight local variations (round small knots). The minimum distance over which the slope of grain is measured shall not be less than the distance over which a deviation of 25 mm is permitted. The slope of grain shall be measured on the face where it is the most severe.
 - Grading Limits: The slope of grain shall not exceed 1 in 10 (30 mm per 300 mm).
- 10) The timber in the piles shall be pressure impregnated with a suitable wood preservative complying with SS 72 at a pressure of not less than 14 kg/cm² and with an average dry salt retention of 24 kg/m³ in the treatable volume of a charge of piles.
- 11) A sample boring taken from any part of any treated pile shall give evidence of continuous and substantially even penetration of the preservative for the full depth of the treated zone.
- 12) A treatment certificate shall be submitted to the S.O. by the Contractor for every consignment of piles delivered. The certificate shall give details of charge number, average dry salt retention and the number and sizes of piles supplied.
- 13) All timber piles delivered to site shall bear the SISIR marking pertaining to the standards attained for tests on timber piles pressure treated with approved chemical preservatives.
- 14) However, the S.O. reserves the right to carry out any further tests he deems necessary.
- 15) Pile shoes to timber piles shall only be used where specified.

(viii) Bakau Piles

- 1) Bakau piles shall be good, sound, straight piles with fresh uninjured bark and of reasonably uniform size throughout. They shall be supplied in 3m, 4.5 or 6m lengths and measured after the heads have been squared and the under-sized ends cut-off. They shall not be less than 100mm in diameter at the larger end and not less than 60mm in diameter at the smaller end, measured within 150mm from HS end, such measurements shall not be over knobs and protuberances. The driving head shall be chamfered off at its edges to minimise splitting during driving, and the lower end shall be pointed. The piles shall be kept either in a totally dry or a totally wet condition at all times.

(c) Installation of Bearing Piles

(i) General - Bearing Piles of all Types

- 1) The Contractor's method of working shall be such that the installation of the piles shall not cause any damage to adjacent structures or foundations.
- 2) The Contractor shall select his method or times of working or both to limit the noise of driving to reasonable levels and to comply with environmental legislation.
- 3) The sequence of working shall be to the approval of the S.O. and shall be arranged so as to minimise the vertical and lateral displacement of completed piles. Levels of the tops of adjacent piles shall be measured at intervals while a pile is being driven. Piles which have risen shall be redriven or forced down to the original resistance.

(ii) Piles shall be constructed within the following tolerances:

- 1) In plan, at the working level of the piling rig, 75 mm in any direction from the position shown on the Drawings;
- 2) 1 in 75 from the vertical for a vertical pile; and
- 3) 1 in 25 from the specified rake for a raking pile.

(iii) Precast Reinforced Concrete Piles

- 1) Piles shall be driven by a hammer of such a type and weight and with such a stroke that no damage is caused to the pile.
- 2) When the 'final set' has been achieved, repeat sets shall be carried out only with the approval of the S.O..
- 3) The driving conditions when taking the 'set' shall be the same as those used when the sets on the non-working piles are obtained.
- 4) The pile head shall be protected during the driving with packing of resilient material evenly spread and held securely in place. A helmet shall be placed over the packing and provided with a dolly of hardwood or other suitable material not thicker than the width of the pile.

- 5) Pile heads shall be stripped down carefully, so as not to damage the pile, to a level such that the remaining concrete will project 50 mm into the pile cap. The reinforcement shall be exposed for a length of 40 times the diameter of the main longitudinal bars, to permit it to be bonded into the cap. Any cracked or defective concrete shall be cut away and replaced with new concrete properly bonded to the old.
 - 6) Where a pile has to be lengthened the concrete at the top of the original pile shall be cut down to expose not less than 200 mm of the longitudinal reinforcement. Additional reinforcement shall be joined by full penetration butt welding in accordance with BS EN 1011-2. Alternatively, the joint may be made by lapping the reinforcement, in which case the reinforcement at the head of the pile shall be exposed for a distance of 40 times the diameter of the bars and the new bars overlapped over this distance.
- (iv) Precast Prestressed Concrete Piles
- 1) The bonding of the pile into the pile cap shall be in accordance with the relevant standard.
 - 2) Any lengthening of piles shall be in accordance with the relevant standard.
- (v) Driven Cast in-place Piles
- 1) Steel casing shall be bottom driven by an internal drop hammer.
 - 2) Concrete casing shall be driven on the pile shoe using a mandrel.
 - 3) If steel casing has to be lengthened joints shall be made by butt welding so that the full strength of the original section is developed. The joints shall be designed to maintain the true alignment and position of the pile section.
 - 4) Concrete casing shall be joined by coating the meeting faces with a plastic jointing compound and covering the closed joint with a steel strap.
 - 5) All joints in the casing and between the casing and shoes shall be watertight during driving and on completion. The casing shall be inspected from the top after installation to check that it is neither damaged nor deformed.
 - 6) The casing shall be completely filled with concrete, every precaution being taken to prevent wasting or the formation of voids.
 - 7) The bonding of the pile into the pile cap shall be in accordance with code requirement.
- (vi) Bored Cast in place Piles
- 1) During boring, the inflow of soil and the formation of cavities in the surrounding ground shall be prevented.
 - 2) Shelling out shall be done with a tool at least 75 mm smaller in diameter than the casing and at such a speed that piping of the soil at the bottom of the boring does not occur.

- 3) Where the hole is dry, concrete shall be poured through a funnel and tube so that the flow is directed and does not hit reinforcement bars or the side of the hole. Where the cut off level is below the groundwater level a pressure shall be maintained on the unset concrete equal to or greater than the groundwater pressure.
- 4) Where drilling fluid is used it is to be such as to:
 - Provide continuous support of the excavation;
 - Be easily displaced during concreting;
 - Not pollute the ground at any stage; and
 - Not coat the reinforcement such as to impair the steel/concrete bond,

Drilling fluid including Bentonite supplied to the site and prior to mixing is to be in accordance with API Specification 13A 5th Edition (1993) Drilling Fluid.

- 5) The bonding of the pile into the pile cap shall be in accordance with code requirement.

(vii) Continuous Flight Augured Piles

- 1) Boring shall be by a continuous flight auger capable of penetrating the ground without drawing surrounding soils into the bore. The auger shall not be raised until the required depth has been reached and until concrete or grout is being pumped down the hollow stem of the auger.
- 2) The rate of withdrawal of the auger shall be so controlled that the concrete or grout forms a continuous monolithic pile having the specified or approved cross section at all levels.
- 3) The reinforcement cage shall be inserted into the concrete with the minimum of delay after completion of concreting or grouting and shall be supported in its correct position with the required projection above the cut-off level.
- 4) The bonding of the pile into the pile cap shall be in accordance with code requirement.

(viii) Steel Bearing Piles

- 1) Piles shall be driven by a hammer of such a type and weight and with such a stroke that no damage is caused to the pile.
- 2) When the 'final set' has been achieved, repeat sets shall be carried out only with the approval of the S.O..
- 3) The driving conditions when taking the 'set' shall be the same as those used when the sets on the test piles, described below are obtained.
- 4) The pile head shall be protected during the driving with packing or resilient material evenly spread and held securely in place. A helmet shall be placed over the packing and provided with a dolly of hardwood or other suitable material not thicker than the width of the pile.

- 5) Piles that have to be lengthened shall be jointed by butt welding in such a manner that the full strength of the original section is developed. The joints shall be designed and constructed to maintain the true alignment and position of the pile sections. The joints in box piles shall be watertight.
 - 6) Where any part of the pile is exposed to the atmosphere or to backfill the thickness of the steel over that required by the design criteria shall be increased on each exposed face by 4 mm.
- (ix) Pressure Impregnated Timber Piles
- 1) Methods of handling and pitching timber piles shall be such that it will not cause undue damage to the piles. A hammer weighing not less than one tonne and dropping not more than 1.2 m shall be used to drive the piles. A driving cap or helmet shall be fitted between the hammer and the pile head and two layers of 20 mm thick plywood or similar material, shall be provided between the head and the helmet and between the helmet and the hammer in accordance with the approved practice.
 - 2) Multi-toothed anti-splitting plates of suitable size shall be fitted to the butt and the toe of each pile to minimise splitting.
 - 3) After driving, the heads of the piles shall be cut off square to sound wood and to required level and treated as follows.
 - 4) The cut-off pile head shall be given two flood coats of approved preservative allowing each application to soak into the end grain completely. Approximately 1 hour shall be allowed between each application.
 - 5) After the flood coats application had dried, the pile head shall be heavily coated with an approved bituminous compound for the whole depth that the pile will be encased in the pile cap.
 - 6) When it is necessary to employ piles from two or more lengths, the butting surfaces shall be cut square to ensure contact over the whole cross section of the pile. These joints shall be formed by using mild steel welded box joint of suitable section, 450 mm long and 5 mm thick. The joints shall be dipped in bituminous paint before use, driven to a tight fit with the pile hammer and spiked to both sections of the pile.
- (x) Bakau Piles
- 1) Bakau piles shall be driven by means of a monkey at least 250 kg in weight suspended from a shear-leg frame or small pile-frame. A suitable steel helmet shall be used to protect the head of the pile from damage during driving.
 - 2) The piles shall be driven below the dry season water table level and shall be evenly spaced over the whole base at centres as indicated on the Drawings or as directed by the S.O..

- 3) Where lengthening is necessary, this shall be accomplished by use of a 6 mm wall thickness galvanised iron pipe collar 450 mm long. The collar shall be provided with four (4) pairs of holes for bolts. Bolts shall be 12 mm diameter mild steel. Bakau piles shall be extended by cutting square and trimming their ends batting them together inside the collar. Bolts shall be driven through the holes provided in both directions at both ends of the collars.
 - 4) After driving is complete, the tops of the piles shall be cut-off to a uniform level, 150 mm above the final excavated level. Piling shall be completed before the final 150 mm of hand excavation is carried out.
 - 5) Where specified in the Specification or indicated on the Drawings, load tests on bakau piles shall be carried out by subjecting a group of the piles to a vertical test load, until actual failure of the group takes place. The group of piles shall consist of at least four (4) bakau piles (i.e. 2 rows of 2 Nos.), and shall be driven at a location of the Site as selected by the S.O. The piles tested shall not afterwards form part of the foundation of any structure. At least 48 hours' notice of the commencement of each test shall be given to the S.O.. The load test shall only commence after the piles have been driven and left in position for a period of not less than 7 days.
 - 6) The load test shall be performed before commencing to drive the remaining bakau piles. From the results of the test(s), the S.O. may direct to vary the number or length of the bakau piles required.
- (xi) Records of Bearing Pile Installation
- 1) The Contractor shall keep records of every pile as follows:
 - Pile type;
 - Pile reference number;
 - Original ground level related to Singapore datum;
 - Nominal diameter or dimensions;
 - Date driven or bored;
 - Date concreted;
 - Depth from ground level to toe of pile;
 - Depth or height from ground level to cut off level of pile;
 - Depth or height from ground level to top of the concrete;
 - Final set for driven piles, weight and drops of hammer;
 - Details of any obstructions observed; and
 - Identification of Supervisor.

- 2) The Contractor shall also take a full driving record for all piles. This full record shall include the number of blows for each 300 mm of driving until the target toe level is approached or until the driving resistance increases to the calculated set value, whichever occurs first; thereafter the number of blows for each 50 mm of driving.
 - 3) All records shall be accurately kept in duplicate as the work proceeds and one copy shall be handed to the S.O. as soon as practicable after driving and not later than the completion of each day's work.
 - 4) On completion of all piling work, the Contractor shall prepare and submit to the S.O. two (2) complete sets of drawings showing details of all piles driven at the Site.
 - 5) All pile test records shall be considered as confidential and shall not be disclosed by the Contractor to any third party without the express permission of the S.O..
- (d) Pile Tests on Bearing Piles
- (i) Introduction
 - 1) General
 - Unless otherwise specified, pile tests on bearing piles shall be carried out as described in this Section of the Specification.
 - 2) Test Load and Kentledge
 - The test load shall be applied by means of a hydraulic jack of adequate capacity reacting against kentledge or anchor piles, the jack shall be fitted with measuring devices capable of registering loads in increments not exceeding 5 tonnes. The hydraulic jack and measuring devices shall be approved by the S.O. before use. Current certificates of calibration for all measuring devices shall be provided to the S.O..
 - The kentledge shall be provided by a heavy mass of dead load, exceeding the maximum test load required, supported centrally over the test pile on a suitably constructed loading platform or by means of anchor piles.
 - The loading platform shall be strongly built and suitable arrangements made to prevent overturning. Concrete or metal blocks, bags of sand, structural steel or water in a tank may be used as kentledge.
 - If anchor piles are used to provide reaction for the jack, these piles shall not be closer to the test pile centre to centre than 1.5 m or 5 pile diameters or equivalent whichever is the larger. Anchor piles, which later form part of the piled foundation to the structure, shall be appropriately redriven to the required set or level as directed by the S.O. all to his full satisfaction.

3) Settlement Measurement

- Vertical movements of the test pile shall be measured by at least two dial gauges, with a minimum of 25 mm travel, graduated in 0.025 mm divisions.
- The gauges shall be erected on stiff steel beams, the end of which shall rest on or be fixed to reliable supports. Such supports shall be located not closer than 1.8 m from the test pile and shall not be rigidly fixed to the ground with a depth of not less than 1 m of concrete surround. In addition, the elevation of the supports shall be checked frequently with reference to a fixed benchmark. The entire measuring assembly shall be properly protected against rain, direct sunlight and other disturbances that might affect its reliability. Temperature readings shall be taken when requested by the S.O..

4) Equipment and Personnel for Load Test

- All kentledge, staging, anchor piles, equipment, shelter, supports, and the like required for the load test shall be provided by the Contractor, who shall also provide all necessary facilities to enable the S.O. to check all readings during the progress of the test.
- The S.O. will supervise and direct the whole process of the test and provide skilled staff to assist in taking and recording all readings.
- The Contractor shall provide all other labour required for the proper execution of the test, including the construction and dismantling of the load staging and handling of the kentledge materials on and off the load stage, and the like and for the continuous and competent watching of the whole assembly, all to the full satisfaction of the S.O..

(ii) Frequency of Testing

- 1) Piles shall not be tested until crushing tests on control cubes from the same mix and cured in the same way as the piles, show strengths greater than the specified works cube strength at 28 days.

• Static Load Testing

The Contractor shall carry out maintained load tests on piles as follows:

- o Non-working piles, constructed for test purposes at positions selected by the S.O., shall be tested to the ultimate bearing capacity. These test piles shall be constructed in the same manner and to the same set (where applicable) as the working piles. The number of non-working piles shall be 1 number or 0.5 per centum (0.5%) of the number of working piles, whichever is greater; and

- o 2 number or 1 per centum (1%) of working piles or 1 for every 50m length of proposed building, whichever is greater, i.e. piles forming part of the Permanent Works, will be selected at random by the S.O. and shall be tested to a load of one and a half times the working load. Load tests shall commence not later than 7 days after the instruction from the S.O..
- (iii) Dynamic Load Testing
 - 1) In addition to the requirements for static load testing, the Contractor shall test 2 numbers or 2 per centum (2%) of Working Piles installed by dynamic testing, whichever is greater.
- (iv) Procedure for Testing
 - 1) Static Load Testing

The 'MAINTAINED LOAD' method of testing shall be carried out on the test piles. The S.O. may at his discretion decide to carry out the 'CONSTANT RATE OF PENETRATION' method of testing on completion of the Maintained Load Test.

 - Maintained Load Test
 - o The test load shall be applied using a jack by a method approved by the S.O. and shall be measured within an accuracy of 2%. Settlement recovery shall be measured within an accuracy of 0.05 mm from a datum that will not itself be affected by any movement of the ground due to the test loading or other causes. The S.O. may require independent checks on the level of the datum using an accurate levelling instrument.
 - o The jack shall be properly and recently calibrated by an approved testing station and the appropriate certificate produced before any tests commence. The measuring equipment and supports employed shall be adequately protected from the sun and elements and readings of the air temperature shall be made and recorded at the same as those on the gauges of the several measuring instruments used.
 - o The test load shall be applied in equal increments of one-tenth of the working load up to the working load and thereafter at 20% of the working load up to twice the working load. Each increment of load shall be applied as smoothly and as expeditiously as possible. Settlement readings and time observations shall be taken before and after each new load increment.
 - o The load shall be kept constant after each increment or reduction.
 - o A settlement-time graph shall be plotted to indicate when the rate of settlement of 0.05 mm in 15 minutes is reached. A further increment of load shall be applied when this rate of settlement is achieved or until a minimum time of 2 hours has elapsed, whichever is later. The process shall be repeated until the test load of twice the working load is reached.

- o The load shall be applied and removed gradually and without shock. Other work that might cause vibration or any other disturbance to the ground conditions at the test site or affect the plant or apparatus being used in the test shall be suspended during the period of the test.
- o The working load shall be maintained for at least 12 hours and the test load of 2 times the working load shall be maintained until the rate of settlement ceases to be apparent but for not less than 48 hours.
- o Time-settlement readings shall be taken in regular intervals as for the other load stages.
- o The amount of settlement and recovery shall be read and recorded immediately after the change of loading and at 10 minute intervals thereafter. Readings under working load, full test load and during recovery at zero load should be taken and recorded at 1 hour intervals. The test load shall then be decreased in four equal stages and time-settlement readings taken as described aforesaid until the movement ceases. At least 60 minutes intervals shall be allowed between the unloading decrements.
- Constant Rate of Penetration Test
 - o The load shall be applied at a constant rate of penetration at a value varying between 0.75 mm per minute to 1.5 mm per minute, the rate to be so chosen as to suit the available load application equipment. Both settlement and time readings shall be recorded every minute. Further loading shall be discontinued when the loading varies directly as the penetration in the case of end-bearing piles in sand or gravel or when the rate of penetration is constant without further increase in the load in the case of friction piles in clay. Loading shall then be released gradually and rebound readings taken.

2) Dynamic Load Testing

- Dynamic load tests shall be carried out on working piles selected by the S.O.. These tests involve monitoring the response of a pile to an impact applied to the pile head. Details of the equipment to be used and of the method of analysis of test results shall be provided to the S.O. before the commencement of the works.
- The Contractor shall supply everything necessary for the performance of the tests, and this shall be deemed to have been included in the Contract Sum.
- The Contractor submit the initial test results, which shall include a copy of the field data obtained during the testing but exclude detailed analysis, to the S.O. within 24 hours of the completion of the tests.

- The Contractor shall examine and interpret the results in detail and prepare a report on the tests. The report shall be submitted to the S.O. within 10 days of completion of the testing and shall contain information in accordance with the following schedule where applicable.
- The following information shall be given where applicable:
 - o Pile reference number;
 - o Pile Type and size;
 - o Pile toe level;
 - o Total length of pile;
 - o Length of instrumentation location from toe;
 - o Working load;
 - o Date of pile installation;
 - o Date of test;
 - o Penetration of pile below commencing surface;
 - o Identification of Supervisor;
 - o Identification of person carrying out the analysis;
 - o Hammer used for testing;
 - o Distance of gauges below pile head;
 - o Blow selected for analysis; and
 - o Plot of force at the location of the instruments against time.
- The test records for dynamic pile tests shall include the following information:
 - o An interpretation of each record;
 - o A review of any anomalies in terms of influence of poor materials or workmanship during pile construction;
 - o Maximum compressive force delivered to the pile at impact;
 - o Maximum percussive energy delivered to the pile;
 - o Hammer operating rate;
 - o Dynamic soil resistance, divided between friction and end resistance;
 - o Static resistance, divided into friction and end resistance, which may be extrapolated to pile bearing capacity; and
 - o Estimates of the settlement characteristics of the pile at 1.0 times working load, 1.5 times working load and maximum test load.

- Any deviation in the record from that which would be expected from a sound pile without defect shall be reported. The report shall indicate the nature, location severity of the defect and recommendations shall be made for further testing. Where possible the implication of the existence of the defect on the performance of the pile shall be evaluated.

3) Test Acceptance Criteria for Working Piles under Static Load Test

- The test on working piles shall be deemed to be unsatisfactory if any of the following conditions apply:
 - o The maximum settlement at the head of the pile during the test exceeds the value

$$\frac{PL}{EA} + \frac{D}{120} + 4mm$$

Where - P is equal to 1.5 times the working load;

- L is the length of the pile in mm;
- A is the cross-section area of the pile in mm²;
- E is the Young's modulus for the material of the pile in kN/mm²; and
- D is the least lateral dimension of the pile in mm;
- o When the rate of recovery after the removal of the maximum test load is less than 0.1 mm/hour observed in a period of not less than 15 minutes, the residual settlement at the head of the pile exceeds the value $D/120 + 4$ mm.
- o The test acceptance criteria for non-working piles shall be as set out in the general specifications.

(v) Action in the Event of Pile Test Failure

- 1) If a non-working pile fails to achieve the specified ultimate bearing capacity, the Contractor shall submit for the approval of the S.O. his alternative proposals for piles to meet the Specification. If in the opinion of the S.O. a test on a working pile indicates that the pile does not have the specified ultimate bearing capacity, the Contractor shall submit for the approval of the S.O. his proposals for (a) further testing of working piles and (b) measures to ensure the integrity of the foundation.

(vi) Procedure for Testing Bakau Piles

- 1) Bakau piles for load testing shall be tested to failure by the maintained load method, as specified in the general specifications. The load on the test piles shall be applied in 6 to 8 equal increments until failure or ultimate load is reached. Ultimate load is the load at which the piles continue to settle at a constant rate without further increase of load.
- 2) For each increment of load, the settlement readings shall be recorded at ten minutes time intervals until the rate of settlement is less than 0.25 mm per hour. The load at each increment shall be maintained at a constant value throughout the test.

- 3) The process of applying the next increment of load and taking the settlement readings against various time intervals shall be repeated until the ultimate load for the pile group is reached. The test-load shall then be gradually released. No further settlement readings or rebound readings need be taken on release of the load.
- 4) Graphs of settlement-time, load-time and load-settlement, to suitable scales, shall be plotted and submitted to the S.O. with all records of the readings.
- 5) Interpretation of the Test Results for Bakau Piles.
- 6) The allowable load on the pile group is obtained by applying an appropriate factor of safety to the ultimate load determined from the load test. The factor of safety is applied chiefly to off-set against the possibility of settlement due to consolidation. A factor of safety from 3 to 6 may be applied. As a guide, a factor of safety of 6 shall be applied to very soft soils and a factor of safety of 3 shall be applied to medium strength soils.

(vii) Records of Tests on Bearing Piles

- 1) The Contractor shall, within 48 hours of the completion of the test, submit to the S.O. for each pile tested a detailed record of testing together with graphs prepared to suitable scale, showing:
 - Load and settlement plotted above and below a common base line of time;
 - Settlement and recovery plotted vertically against a baseline of load; and
 - The Contractor shall submit two (2) copies of all records of results, not including the original copy, together with load-settlement and load-time graphs to the S.O. on completion of each test.

(e) Preliminary Test Piles

(i) General

- 1) Where preliminary test piles are specified by the S.O., the Contractor shall include in his Tender a price for the installation and test loading of preliminary test piles prior to the installation of the working piles. The price shall include the supply of all necessary labour, materials, plant and equipment to install the piles, to apply the load and to measure the settlements all in the manner described. The Contractor will be allowed to maintain his equipment on site after forming these piles at his own expense but no guarantee will be given of the time elapsing between completion of the test and commencement of the working piles.
- 2) The Contractor shall give the S.O. at least 48 hours' notice of the commencement of construction of any preliminary pile which is to be test-loaded.

- 3) The location of the test piles shall be specified by the S.O.. Any variation will be permitted only with prior approval of the S.O.. Extra reinforcement and concrete of increased strength may be permitted in the shafts of preliminary piles at the discretion of the S.O.. Prior to the installation of piles driven for preliminary load tests, the Contractor shall submit to the S.O. detailed calculations substantiating the pile design that he proposes to adopt to achieve the specified ultimate load carrying capacities of the piles.
 - 4) For each preliminary pile which is to be tested, a detailed record of the conditions experienced during boring, or of the progress during driving, shall be made and submitted to the S.O. daily, not later than noon on the next working day. Where the S.O. requires soil samples to be taken or in situ tests to be made, the Contractor shall present the results without delay.
 - 5) Four test cubes shall be made from the concrete used in the preliminary test pile and from the concrete used for building up a working pile. If a concrete pile is extended or capped for the purposes of testing, a further four cubes shall be made from the corresponding batch of concrete. The cubes shall be made and tested.
 - 6) The pile test shall not be started until at least 7 days have elapsed after completion of the pile installation or until the strength of the cubes taken from the pile exceeds twice the average direct stress in any pile section under the maximum required test load, and the strength of the cubes taken from the cap exceeds twice the average stress at any point in the cap under the same load whichever is later.
- (ii) Test Pile Cut-Off Levels
- 1) The cut-off levels for preliminary test piles shall be approved by the S.O..
 - 2) Where the cut-off levels of preliminary test piles and working piles are below the ground level at the time of pile installation and where it is required to carry out load tests from that installation level, these tests piles shall be sleeved from ground level down to the cut-off level of working piles at the pile test locations, such that during testing the pile shaft is completely free of soil contact and skin friction above the proposed cut-off level. The Contractor shall submit with his tender a method statement and sketches showing the procedure he proposes to adopt to comply with this requirement.
- (f) Safety Precautions in Load Tests
- (i) The Contractor shall ensure that the design, erection and dismantling of the pile test reaction system and the application of load shall be carried out according to the requirements of the various applicable statutory regulations concerned with lifting and handling heavy equipment and shall safeguard operatives and others who may from time to time be in the vicinity of a test from all avoidable hazards.

- (ii) Where kentledge is used the Contractor shall construct the foundations for the kentledge and any cribwork, beams or other supporting structure in such a manner that there will not be differential settlement, bending or deflection of an amount that constitutes hazard to safety or impairs the efficiency of the operation. The kentledge shall be adequately bonded, tied or otherwise held together to prevent it becoming unstable because of deflection of the supports or for any other reason.
- (iii) The weight of kentledge for each test shall be greater than the maximum test load for that test, and if the weight is estimated from the density and volume of the constituent materials an adequate factor of safety against error shall be allowed. Additional kentledge required shall be determined taking into account the accuracy of positioning of the centre of gravity of the stack.
- (iv) Where tension piles, reaction piles or ground anchorages are required to provide the necessary load reaction, they shall be so designed that they will resist the forces applied to them safely and without excessive deformation which could cause a safety hazard during the work. Such piles (which unless approved will not be working piles) or anchorages shall be placed in the specified positions, and bars, tendons or links shall be aligned to give a stable reaction in the direction required. Any welding employed to extend or to fix anchorages to a reaction frame shall be carried out so that the full strength of the system is adequate and unimpaired.
- (v) In all cases the Contractor shall ensure that when the hydraulic jack and load-measuring device are mounted on the pile head the whole system will be stable up to the maximum load to be applied.
- (vi) If in the course of carrying out a test any unforeseen occurrence should take place, further loading shall not be applied until a proper S.O. assessment of the condition has been made and steps have been taken to rectify any fault. Reading of gauges should, however, be continued where possible and if it is safe to do so.
- (vii) Where an inadequacy in any part of the system might constitute a hazard, the Contractor shall provide means to enable the test to be controlled from a position clear of the kentledge stack or test frame.
- (viii) The hydraulic jack, pump, hoses, pipes, couplings and other apparatus to be operated under hydraulic pressure shall be capable of withstanding a pressure of 1.5 times the maximum pressure used in the test without leaking.
- (ix) The maximum test load expressed as a reading on the gauge in use shall be displayed and all operators shall be made aware of his limit.
- (x) For a pile that is tested in compression, the pile head or cap shall be formed to give a plane surface which is normal to the axis of the pile, sufficiently large to accommodate the loading and settlement measuring equipment and adequately reinforced or protected to prevent damage from the concentrated application of load from the loading equipment.
- (xi) Any pile cap shall be concentric with the test pile; the joint between the cap and the pile shall have a strength equivalent to that of pile.

- (xii) Sufficient clear space shall be made under any part of the cap projecting beyond the section of the pile so that, at the maximum expected settlement, load is not transmitted to the ground by the cap.
 - (xiii) For a pile that is tested in tension, means shall be provided for transmitting the test load axially without inducing moment in the pile. The connection between the pile and the loading equipment shall be constructed in such a manner as to provide strength equal to the maximum load which is to be applied to the pile during the test, with an appropriate factor of safety on the structural design.
- (g) Method of Load Testing Piles
- (i) The setting-up of pile testing equipment shall be carried out under competent supervision and the equipment shall be checked to ensure that the setting-up is satisfactory before the commencement of load application.
 - (ii) All tests shall be carried out only under the direction of an experienced and competent supervisor conversant with the test equipment and test procedure. All personnel operating the test equipment shall have been trained in its use.
 - (iii) The test load shall be applied by means of a hydraulic jack through a load cell or proving ring newly calibrated in divisions not exceeding 2% of the maximum load applied. A certificate of calibration shall be provided at the S.O.'s request.
 - (iv) Where kentledge is used for loading vertical piles in compression, the distance from the edge of the test pile to the nearest part of the crib supporting the kentledge stack in contact with the ground shall be not less than 1.3m.
 - (v) The centre-to-centre spacing of vertical reaction piles, including working piles used as reaction piles, from a test pile shall not be less than three times the diameter of the test pile or the reaction piles or 2m, whichever is the greatest, except in the case of piles of 300mm diameter (or equivalent) or less, in which case the latter dimension may be reduced to 1.5m.
 - (vi) If ground anchors are used to provide reaction for the jack, the centres of these anchors shall be located least 2.0 metres from the test pile centre or three times the diameter of the test pile, whichever is the greater. The minimum period between installation and test load of a pile shall be one week.
 - (vii) The reaction frame support system shall be adequate to transmit the maximum test load in a safe manner without excessive movement or influence on the test pile. The Contractor shall submit calculations to the S.O. when required to justify the design of the reaction system.

- (viii) Throughout all tests observations of vertical movement shall be made with two dial gauges graduated in 0.025mm divisions. A linear metric scale, graduated with the appropriate number of sub-divisions shall be placed vertically on either side of the test pile to facilitate the measurement of its vertical movement. The readings shall be referred to two stiff steel reference beams the ends of which rest on, or are fixed to, reliable supports. Such supports shall extend not less than one metre below ground surface and shall not be located closer than three test pile diameters or two metres from the centre of the test pile or ground anchors where used whichever distance is the greater. In addition, the elevation of the supports shall be checked frequently with reference to a fixed benchmark approved by the S.O.. The entire measuring assembly shall be protected against rain, direct sunlight and other disturbances that might affect its reliability.
- (ix) At least 48 hours' notice of the commencement of each test shall be given to the S.O..
- (x) The Contractor shall keep the test under continuous and competent supervision to the satisfaction of the S.O. and he shall send two signed copies of all records and graphs to the S.O. within three working days of the completion of each test. Graphs shall be plotted to a suitable scale showing:
 - 1) Load and settlement plotted above or below a common base line of time;
 - 2) Load only plotted vertically above a base line of settlement;
 - 3) All necessary facilities shall be provided to enable the S.O. to check readings during the progress to the test;
 - 4) Load testing shall be of the Incremental Loading Method. Loading shall be in increments of 25% Working Load to 2.0 times that load and then where further loading is required in similar or smaller steps as directed by the S.O.;
 - 5) Unloading shall be in at least three equal steps. During the loading cycles, 1 time and 2.0 times Working Load shall be maintained for a minimum of 24 hours or until settlement does not exceed 0.1mm in an hour. All other loading steps including maximum load and zero load at the end of the test shall be observed for a period of not less than one hour and until settlement or rebound does not exceed 0.1mm in 20 minutes. The final rebound shall be recorded 24 hours after the entire test load has been removed;
 - 6) The exact tonnage for such increment or decrement of loading shall be determined by the S.O.. The period between increments and decrements of loading may be prolonged or shortened at the discretion of the S.O. who shall take into consideration the behaviour of the pile under test; and
 - 7) Sufficient readings for such load steps shall be taken to define properly all the time-settlement and the time rebound curves. The elevation of the datum beam supports and the test pile shall be frequently checked by precise level with reference to a fixed benchmark.

(h) Load Testing Preliminary Test Piles

- (i) The test pile shall first be loaded to 2.0 times the Working Load by Incremental Loading with the load maintained for a minimum of 24 hours or until settlement does not exceed 0.1mm in an hour, and then unloaded in at least three equal steps.
- (ii) After completion of this loading and unloading cycle, the test pile shall then be loaded by Incremental Loading to 3 times or 2.5 times the Working Load for bored piles and driven piles respectively with that load maintained for a minimum of 24 hours or until settlement does not exceed 0.1mm in an hour, or to failure whichever is the lower.
- (iii) The failure load is defined as the constant load at which the test pile continues to settle at a steady rate or the load at which the total settlement of the pile top is one tenth of the pile base diameter or 25mm, whichever is the lower. The increment of loads immediately before and after the estimated ultimate/failure load shall be reduced to define more accurately the load settlement curve to determine the actual ultimate/failure load of the pile.

(i) Interpretation of Preliminary Test Pile Results

- (i) The performance of the test piles under load test will be used to determine the bearing capacities of the working piles.

After consideration of the results from the preliminary test piles, the S.O. will take action as follows:

- 1) The Contractor will be instructed in writing to proceed with the installation of the working piles as scheduled, or
- 2) Revised sizes of working piles will be agreed with the Contractor without further testing before work proceeds, or
- 3) The Contractor will be instructed to place and test additional test piles.
- (ii) In case of failure of Preliminary Test Piles, the Contractor shall not be paid for additional work mentioned in this Clause unless the cause of failure does not arise out of the Contractor's negligence, incompetence or failure to comply with the conditions stipulated by this Specification.

(j) Load Testing of Working Piles

- (i) The S.O. will select piles to be tested after their installation. Selected piles shall not be loaded until at least one week after their installation or until the pile concrete has reached its specified 28-day concrete cube compressive strength whichever is the longer.
- (ii) The use of working piles as tension piles for the purposes of these tests will not be permitted.
- (iii) The Contractor shall allow for extending the pile and the casting of any pile cap as required by the S.O. for the testing to be carried out. Pile caps for the load tests, if used, shall be designed and constructed by the Contractor to the approval of the S.O.. The cost of such pile caps shall be deemed to have been included in the Contract Sum.

- (k) Allowable Settlement under Pile Load Test
 - (i) The working pile load test for which the pile is usually tested to 1.5 to 2.0 times working load, the allowable maximum settlement measured at the top under full test load is generally taken as 15mm or 25mm respectively. In the case where elastic shortening of the pile is large at full test load, such as when the test involves a very long steel pile the allowable maximum pile top settlement may be increased.
 - (ii) For piles subject to negative skin friction, the working load test can be performed in the normal manner, but with its acceptance subject to satisfaction of an additional criterion that pile settlement at the test load of 1.0 times column load plus 2.0 times negative friction should not exceed 10mm.
- (l) Failure of Working Pile Tests
 - (i) A pile or pile group shall be deemed to have failed if any of the following observations is recorded:
 - 1) The maximum settlement under full test load exceeds the allowable settlement limits stated above;
 - 2) The permanent or residual settlement after removal of the test load exceeds 5mm for piles with nominal working load less than or equal to 100 tonnes, or 0.05mm per tonne of nominal working load for piles with nominal working load greater than 100 tonnes subject to a maximum of 13mm;
 - 3) Failure of the pile materials (due to defects in the pile); and
 - 4) Failure of concrete to reach the design compressive strength.
 - (ii) In the event of one or more working piles fail to satisfy the test requirements, the Contractor shall at his own cost test such additional piles as may be required by the S.O.. The Contractor shall also bear the cost of all additional work including the installation of additional piles that the S.O. may consider necessary due to the failure of one or more piles to satisfy the requirements of the loading test.
 - (iii) If, in opinion of the S.O. it is impractical or inadvisable to install substitute piles in place of a failed pile, the Contractor shall submit proposals to rectify the defect. These proposals are subject to the approval of the S.O..
 - (iv) In addition, the Contractor shall carry out two more similar tests. The piles for these tests shall be selected by the S.O. When the additional tests are successful, only one (1) load test will be paid and no payment will be made to the Contractor for the failed test. In case of further failure, the Contractor shall at his own expense conduct further tests and/or proposed remedy as approved by the S.O..
- (m) Abandonment of Pile Test
 - (i) A load test shall be abandoned if it has to be discontinued due to:
 - 1) Pre-jacking or pre-loading before the commencement of the test;
 - 2) Improper setting of datum;

- 3) Disturbed or unstable bench marks or scales;
 - 4) The scales and/or measuring instruments used are found to be defective or have been tampered with;
 - 5) Faulty jack or gauges;
 - 6) Failure of pile cap or instability of the kentledge;
 - 7) Failure of the pile materials; and
 - 8) Failure of concrete to reach the design compressive strength;
- (ii) The cost of the abandoned test shall be borne by the Contractor and he shall carry out one further test to the S.O.'s direction.
- (n) Tension Pile Tests
- This test method is similar to compression testing of piles, and maintained load procedures as described in Clauses G3.13.8 (d) Bearing piles - Pile Tests on Bearing Piles may be used. In the ultimate load test, the test pile shall be loaded to 3 times the Working Load or to failure whichever is the lower. The reaction system may be provided by a strong-back resting on adjacent anchor piles or spread foundations. Two jacking points on either side of the pile are used to avoid bending the test pile, and the Contractor shall exercise due care to ensure that the jacks are loaded evenly.
- (o) Lateral Load Tests
- (i) If the applied lateral loads are small, it may be convenient to use an adjoining pile as a reaction. The Contractor shall provide a stable datum for deformation measurement as for vertical tests. If two identical piles are being tested, it may be more convenient to measure the relative movement between them using an extensometer.
- (ii) In lateral load tests, it is desirable to measure the rotation of the pile head. The Contractor shall accomplish this by using a sensitive spirit level mounted on the pile, or by using an electro-level.
- (p) Rejected Piles and Compensating Piles
- (i) Any pile which falls in the following categories shall be rejected and replaced or compensated:
- 1) Pile which is cracked, deformed, twisted, broken or damaged in any manner;
 - 2) Pile installed out of position or plumb exceeding the tolerable limits specified;
 - 3) Its representative load test fails;
 - 4) Its representative testing on material fails; and
 - 5) Driven pile which fails to achieve the agreed values of set and temporary compression (Refer to Clause G3.13.8 Bearing Piles).

- (ii) All installed piles that are rejected shall be replaced with compensating pile/s at the Contractor's expense. The Contractor shall submit his design proposing the position and number of compensating pile/s that he uses to replace the rejected pile/s for approval by the S.O.. The Contractor's design so proposed shall be such that the centre of the original pile group is not altered. Only the cost for the original number of piles in the pile group as shown on the Drawings will be paid.
 - (iii) In addition, the Contractor shall bear all the extra costs occasioned by such rejection due to the redesign and construction of increased size of pile caps, strengthening of tie beams, etc.
- (q) Removal of Obstruction
 - (i) The Contractor shall be deemed to have included and allowed for in his Tender excavation in any material to remove obstructions whether informed or unexpected and which prevent the installation of the piles.
 - (ii) No claims for standing time for plant, manpower, etc. shall be allowed for in consequence of obstructions of any kind.
 - (iii) Removal of interlocking concrete panels at the canal embankment and canal bottom slab may be required during piling. The contractor shall reinstate the canal embankment and bed to original condition once piling has been completed.
- (r) Sequence of Piling
 - (i) Piles shall be installed in an approved sequence to minimise any detrimental effects of heave and lateral displacement of the ground and the Contractor shall ensure that no damage is likely to be caused to piles previously formed at adjacent positions.
 - (ii) The Contractor shall first obtain the S.O.'s approval for his proposed sequence of installation of piles in group or in close proximity before implementation. The S.O. reserves the right to order changes to the approved sequence of piling if, based on work completed, he considers it necessary to do so. The Contractor shall not be entitled to claim extra payment for additional plant movements, etc. resulting from Ultimate Load Testing of Piles.
 - (iii) The Contractor shall be required to provide test piles for ultimate load tests where specified in the Drawings or as required by the S.O. before the installation of the working piles. The test piles, to be installed separately from the working piles and in a position selected by the S.O., shall be of the same material and sizes as the working piles and driven with the same plant.
 - (iv) The driving of the test piles shall be done in a manner and to a final depth and/or set as approved by the S.O. who may order the interruption of the driving for a sufficient length of time to enable the soil to recover from the disturbance of driving. The driving shall be resumed when ordered, to determine the variation in resistance. The Contractor shall be deemed to have allowed for the cost of this interruption as this requirement is part of the driving of test piles.

- (v) The performance of the test piles under driving and later under test load will be used to determine or confirm the bearing capacities of the working piles and driving criteria. The load test on preliminary test pile shall not be implemented until at least 7 days have elapsed after driving has been completed. The time required for the installation and testing of test piles shall be deemed to have been included in the Contract Period.
 - (vi) The ultimate load test shall be used to verify that the ultimate load of the piles is at least 3 times the design working load.
 - (vii) Based on the results of the tests, the S.O. may require the Contractor to modify his design capacity assumed for the piles. The Contractor shall not be entitled to any extra payment as a result of such change.
 - (viii) The ultimate load shall be the load at which the pile continues to settle or heave without further additional loading. For practical purpose it shall be taken as that load which causes the head of the pile to settle or heave an amount equal to 10% of the effective pile diameter unless the value of the ultimate bearing capacity is defined clearly in the load/settlement or load/uplift curve. The effective pile diameter is defined as the diameter of a circle inscribed in the section of the pile. In the case of H section steel piles, it shall be the dimension of the web or flange whichever is smaller.
- (s) Dynamic Load Testing of Piles
- (i) Dynamic pile testing of piles may be used to evaluate the pile capacity, soil resistance distribution, immediate settlement characteristics, hammer transfer energy and pile stresses during driving.
 - (ii) If so directed by the S.O., the Contractor shall engage an independent specialist sub-contractor, to the approval of the S.O., to perform the dynamic pile testing. The dynamic pile-testing and particularly the analysis of test data must be carried out by trained and experienced S.O.. The Contractor shall provide all the necessary attendance for the satisfactory execution of the testing.
 - (iii) Where it is required to evaluate pile driving stresses, testing must be carried out during the installation procedure. Where bearing and resistance evaluation is required, the testing should be carried out sometime after the completion of the pile installation as in the case of static load testing. The dynamic pile testing should not be relied on as the sole means of assessing the acceptability of piles, except in areas where the geology is similar and extensive experience has been accumulated. The dynamic methods should be correlated with and calibrated against the static load test results, before extensive testing of working piles is undertaken.
 - (iv) The Contractor shall submit to the S.O. the results of the dynamic pile testing, including all records, analysis of test data, graphs and calculations not later than 72 hours after completion of the test.
 - (v) The Contractor shall include with his tender the name of the specialist sub-contractor he would propose to engage for the dynamic pile testing, together with full details of the proposed method of testing.

- (t) Final Survey and As-Constructed Drawings and Calculations
 - (i) On completion of the excavation works to the cut-off levels of the piles, the Contractor will be required to employ a registered land surveyor to prepare for the S.O. a survey plan showing the positions, top levels and penetration lengths of the piles, and eccentricities of the piles at cut-off levels; and as-constructed drawings showing vertical inclination of the piles in orthogonal directions measured at the ground and cut-off levels the trimmed levels of the ground and piles and founding levels.
 - (ii) Other information such as the sets and temporary compressions of driven piles shall also be shown on the as-constructed drawings. After these as-constructed drawings have been reviewed by the S.O., the Contractor shall submit three paper prints and CD of each of the as-constructed piling drawing to the S.O. for his record.
 - (iii) Final as – constructed drawings shall be prepared and submitted by the Contractor not more than 2 weeks after the survey has been carried out for the piling works.
 - (iv) The satisfactory submission and acceptance of as – constructed drawings shall be condition precedent to the issuance of completion certificates for the works by the S.O. and the Contractor shall take this into consideration in his scheduling of the Works and drawing preparations.
 - (v) The contractor should keep a set of piling record and calculations on site and submit to the S.O. as and when instructed to do so by the S.O.. All calculations should be based on the design soil parameters approved by the S.O. and attained the prescribed pile capacity shown in the drawings or approved by S.O..

G3.13.9 Concrete Piles

- (a) General

Unless stated otherwise or specifically mentioned in the ensuing clauses in this Section, the Contractor shall comply with all the standards of quality of materials and workmanship as specified in the Specification for Concrete and Steel respectively.
- (b) Cement Replacement Materials

The use of cement replacement materials by the Contractor may be permitted subject to the approval of the S.O. and provided that they can be shown to have beneficial effects on concrete impermeability, heat generation during setting and general durability. The proposed mix proportions inclusive of the cement replacement materials shall be subject to approval by the S.O..
- (c) Designed Mix

Designed mixes shall be a minimum of grade 35 (for cast-in-place piles) or grades 45, 50, 55 or 60 (for precast piles) complying with BS EN 206-1. Other grades may be approved by the S.O. if appropriate to the work. The Contractor shall submit complete information on the mix and sources of aggregates for each grade of concrete, the water/cement ratio and the proposed degree of workability for approval of the S.O. before work commences.

(d) Minimum Cement Content

The cement content in any mix shall be not less than 300kg/m³. Where concrete is to be placed under water or drilling mud by termite or where the pile will be exposed to sea water, the cement content shall be not less than 400kg/m³.

(e) Concrete Cover to Reinforcement

The cover to all reinforcement in piles cast directly against the earth shall be not less than 75mm. Spacers shall be designed and manufactured using durable materials which will not lead to corrosion of the reinforcement or spalling of the concrete cover. For precast piles, the cover shall be in accordance with the requirements of SS CP 65.

(f) Cutting Off Pile Heads

The Contractor shall exercise due care to ensure that no shattering or damage is caused to the rest of the pile in the course of cutting the pile head. The Contractor shall submit his method statement for the hacking of the pile heads for approval by the S.O. prior to commencement of Works on Site. Any cracked or defective concrete shall be cut away and the pile repaired in an approved manner to provide a full and sound section to cut-off level.

G3.13.10 Bored Piles

(a) Method of Bored Pile Excavation

(i) The excavation for bored piling shall be carried out generally using boring equipment operating at the existing ground level in accordance with the relevant requirements of SS EN 1992-1-1. During boring, the Contractor shall collect soil/rock samples from the boring and submit them in suitably labelled and approved air-tight containers to the S.O. for examination and testing as and when required.

(ii) Installation of bored piles shall be carried out by experienced piling specialist. Prior to piling, proper survey and marking out of the locations of the bored piles be carried out by a registered Land Surveyor. Before commencement of piling, the Contractor shall engage a carry out trial trenching to confirm the proposed pile locations are clear of existing piles and services.

(b) Boring Near Recently Cast Piles

No Pile shall be bored or pile casing be driven nearer than six times larger diameter (measured center to center) from an unfilled pile borehole or from an uncased concrete pile where the concrete had been placed less than 24 hours previously.

(c) Obstruction

(i) If the proposed pile clashes with the existing pile, the Contractor shall propose adjustments to the proposed pile location that will not change the centre of gravity of the original pile group.

(ii) If such adjustment affects the pilecap design, the contractor's PE shall propose remedial actions for S.O.'s approval before carrying out these actions. The Contractor shall also check and ensure the verticality of the pile is within 1H: 75V. The Contractor shall use steel casings for the installation of bored piles.

- (iii) In the event of obstructions below ground level being encountered during boring, the Contractor shall immediately inform the S.O.. If in the opinion of the S.O. further progress on that pile is impractical by the method employed, the S.O. shall instruct the Contractor for the necessary action to be taken.
 - (iv) Where the S.O. instructs the Contractor to abandon the pile from this cause and proceed with piling work on another pile, the Contractor will be paid only for the length of empty boring and backfilling.
- (d) Bentonite
 - (i) Bentonite for use in the support of pile bore where its usage has been approved shall be in accordance with the latest edition of Specification No. D.F.C.P.4 of the Oil Companies Materials Association, London. Samples of the bentonite to be used by the Contractor, together with manufacturer's details shall be submitted to the S.O. for approval.
 - (ii) A certificate shall be provided for the S.O. with each consignment stating the properties of samples tested. Samples shall be taken so that at least one set of tests is carried out for each consignment of bentonite delivered to the Site and not less than one for every ten tonnes of bentonite supplied. Tests shall be carried out and the following information provided:
 - 1) Apparent viscosity and gel strength of a suspension of bentonite in water; and
 - 2) Moisture content of the bentonite powder and wet screens residue on US200 mesh sieve (74µm).
 - (iii) A statement of the conditions of test, such as water analysis, mixer type, mixer speed during testing, timing of stirring and time of testing after initial addition of bentonite to the water.
 - (iv) For any 100 tonnes of bentonite, moisture content shall not deviate more than plus or minus 2%, apparent viscosity by more than plus or minus 2cps and gel strength measured at 10 minutes by more than plus or minus 0.023N/sq.m.
 - (v) Allowance shall be made for one sample of bentonite powder per three pile bores to be tested for its apparent viscosity, gel strength, moisture content and wet screen residue under the same conditions as those under which testing was carried out by the supplier or manufacturer, at the date that work on the pile bores is commenced.
- (e) Mixing of Bentonite Slurry
 - (i) Water used for dispersion of bentonite shall be as nearly neutral in pH as practicable and care shall be taken to exclude saline water or water contaminated by salts.
 - (ii) Bentonite powder shall be mixed thoroughly into the water by a suitable high shearing action mixer until all lumps have been broken up and dispersed within the mix.

- (iii) The proportion of bentonite to be included in the slurry shall take into account the rheological properties of the bentonite such that the slurry shall be sufficient to maintain the stability of the pile bore for the period necessary to place concrete and complete construction. The density and head of slurry above ground water shall together be such as to ensure the stability of the pile bore throughout its entire depth.
- (iv) Where saline or chemically contaminated groundwater occurs, the Contractor shall take special precautions to modify the bentonite suspension or prehydrate the bentonite in fresh water so as to render it suitable in all respects for the construction of piles.
- (f) Testing of Bentonite Slurry
 - (i) The test equipment and procedures employed shall be in general as specified in American Petroleum Institute Standards (Ref. API-RP-13B-1 & 13B-2). Additional or alternative equipment and procedures may be employed with the approval of the S.O..
 - (ii) The bentonite slurry in any pile bore during and after excavation and immediately before and during concreting shall be tested for density using a mud balance, viscosity, yield stress and gel strength using a Fann viscometer, viscosity on a Marsh funnel, sand content, fluid loss, pH using an electric pH meter and temperature. At the discretion of the S.O.. Fann viscometer readings may be discontinued when consistent results have been established.
 - (iii) Density measurements shall take into account the moisture content of the bentonite powder in determining the bentonite concentration.
 - (iv) pH shall be monitored during the addition of additives.
 - (v) A pattern of relationship between gel strength, viscosity and density shall be established on Site so that:
 - 1) The gel strength/viscosity relationship shall not produce excessive combined values likely to produce slurry entrapment;
 - 2) No flocculation of the bentonite occurs; and
 - 3) No settlement of the solids within the slurry occurs.
 - (vi) Allowance shall be made for taking samples of bentonite slurry from the pile bore as follows:
 - 1) One sample per 5m depth or one sample per each change in soil type, as determined by the S.O. during boring of each pile position;
 - 2) Two samples (or more, as necessary to demonstrate compliance with the specification), from the bottom of the pile bore prior to placing the reinforcement cage, and likewise immediately before concreting; and
 - 3) Three samples from near the top of the concrete during concreting of each pile bore.
 - (vii) The range of values for slurry properties shall conform to that given in the following table unless otherwise agreed with the S.O.:

	During Excavation	Before & During Concreting
Density (gms/cc)	1.02-1.08	1.02-1.06
Marsh funnel viscosity (seconds)	33-42	33-42
Apparent viscosity (centipoise)	5-12	5-12
Plastic viscosity (centipoise)	10-15	10-15
Bingham Yield (lbs/100 sq.ft)	5-20	5-20
Gel strength-10 minute (lbs/100 sq.ft)	3-8	3-8
Sand content (%)	5	4
Fluid loss (cc)	35	35
PH	7-12	8.5-11.5

- (viii) The S.O. may require the slurry to be changed if, in his opinion, a sample obtained from as near the bottom of the pile bore as possible before cage placement shows that such treatment is necessary for concrete placement without undue contamination by bentonite.
- (g) Site Laboratory for Testing Bentonite / Bentonite Slurry
 - (i) The Contractor shall establish a suitably equipped and properly maintained site laboratory and provide skilled staff and all necessary apparatus to undertake all the sampling and testing stipulated in this General Specification.
 - (ii) The accuracy of the testing equipment shall be checked and calibrated by an approved independent testing agency.
- (h) Storage of Bentonite
 - (i) Bentonite shall be stored in dry cool conditions. Particular care shall be taken with bulk storage to prevent balling together of bentonite powder due to dampness or deterioration of properties due to dampness and heat. A suitable design of hopper cone and bentonite feeding device shall be submitted for approval by the S.O..
- (i) Spillage and Disposal of Bentonite Slurry
 - (i) The Contractor shall take all necessary precautions to prevent the spillage of bentonite slurry on the Site in areas outside the immediate vicinity of the borings.
 - (ii) Discarded bentonite slurry shall be removed from the Site without undue delay.
 - (iii) The Contractor shall comply with the regulations of the local Authorities in disposing the bentonite slurry.
- (j) Temporary Casings
 - (i) Temporary casings shall be used where necessary, or where directed by the S.O. to prevent the inflow of soil and water, to maintain the pile excavation to its full dimension and to ensure that the pile shaft shall be formed to its full cross-sectional area. In water bearing ground, full length casings shall be used to the founding level. The Contractor shall ensure that the cutting edge of the casing is driven below the base of the advancing bore to prevent the formation of cavities or over breaks outside the nominal diameter of the pile.

- (ii) Temporary casing of approved quality shall be used to maintain the stability of a pile bore which might otherwise collapse.
 - (iii) Temporary casings shall be free from significant distortion. They shall be of uniform cross-section throughout each continuous length. During concreting they shall be free from internal projections and encrusted concrete which might adversely affect the proper formation of piles.
 - (iv) The Contractor shall ensure that the temporary casings are installed in such a manner that they project above the ground at the top of the pile bores to prevent debris and other foreign objects from falling down the pile bores. Likewise, the Contractor shall cover the empty pile bores outside the working periods for the same reason.
 - (v) The method of installing and withdrawing the temporary casing shall not cause disturbance of the ground which would adversely affect the construction or the capacity of piles. The casings must be removed in such a manner that no defect or damage will be inflicted on the newly cast pile. Where the concrete in contact with the casing has passed the setting time, the casing shall be abandoned in the pile.
 - (vi) Where piles are bored under water or bentonite suspension in an unlined state, the insertion of a full-length loosely fitting casing to the bottom of the bore prior to placing concrete will not be permitted.
 - (vii) Where permanent casing is used by the Contractor to ensure the integrity of a pile the Contractor shall submit for approval his proposals regarding the method of installation.
- (k) Stability of Pile Bores
- (i) The Contractor shall take all necessary precautions to ensure stability of the pile bores. Where boring takes place through unstable water bearing strata, the process of excavation and the depth of temporary casing employed shall be such that soil from outside the area of the pile is not drawn into the pile section and cavities are not created outside the temporary casing as it is advanced.
 - (ii) Where the use of bentonite slurry is specified or approved for maintaining the stability of a bore, an adequate temporary casing shall be used in conjunction with the method so as to ensure stability of the strata near ground level until concrete has been placed. During construction, the level of bentonite slurry in the pile excavation shall be maintained within the cased or stable bore so that it is not less than 1.0m above the level of external standing groundwater at all times.
 - (iii) The Contractor shall maintain, available for immediate use, a sufficient quantity of slurry to allow for any sudden loss. In the event of a rapid loss of bentonite slurry from a pile excavation, the bore should be topped up with bentonite slurry without delay. Should the loss continue despite the addition of the slurry and the stability of the pile bore is placed at risk, the Contractor shall immediately advise the S.O. and take such immediate remedial action as is necessary to preserve the stability of the pile bore and ensure the safety of neighbouring structures and services. The Contractor shall obtain instructions from the S.O. before boring at that location is resumed.

- (l) Pumping from Pile Bores
 - (i) The Contractor is not permitted to pump from pile bores unless the bore has been sealed against further water entry by casing or unless the soil is stable and will allow pumping to take place without ground disturbance below or around the pile.
- (m) Continuity of Construction
 - (i) For a pile constructed in a stable cohesive soil without the use of temporary casing or other form of support, the Contractor shall bore the pile and place the concrete without such delay as would lead to significant impairment of the soil strength.
- (n) Inspection
 - (i) Each pile bore shall be carefully inspected on completion by the Contractor and the S.O. before concreting is carried out. Each pile bore which does not contain standing water or bentonite slurry shall be inspected directly or indirectly prior to concrete being placed in it. This inspection shall be carried out from the ground surface in the case of piles of less than 750mm diameter. Torches or other approved means of lighting, measuring tapes, and a means of measuring verticality shall be provided. For piles of 750mm diameter or larger, the Contractor shall propose his method of inspecting the pile bores by his representatives and the S.O.. Any approved equipment used shall be subjected to the approval of the S.O. or his representative.
- (o) Cleanliness of Pile Bases
 - (i) On completion of boring and where inspection of a dry pile bore indicate the necessity, the Contractor shall remove all loose, disturbed or softened soil from the bore. For piles deriving their bearing capacity from a rock socket, the final cleaning of the pile bases may be effected by the use of an air-lift or by adopting the reversed-circulation drilling technique or other approved means.
 - (ii) Where pile bores contain water or bentonite slurry, the base cleaning may be effected by means of a cleaning bucket or with an air lift or other approved means. As fine sand and silt suspended in the bentonite slurry will gradually settle to the bases of the piles, the Contractor shall conduct the final cleaning of the holes shortly before concreting.
 - (iii) The Contractor shall submit the details of the cleaning equipment and his proposed method of removing large debris and/or accumulated sediment from the pile bases for approval by the S.O..
 - (iv) The Contractor shall ensure that the base cleaning will not loosen the pile bases and to cause ground disturbance below the pile bases. The Contractor shall maintain bentonite slurry at such levels throughout and following the cleaning operation that stability of the bore is preserved.
- (p) Reinforcement
 - (i) Reinforcement shall be provided for the piles as detailed on the Drawings. The minimum cover shall be 75mm. The Contractor shall submit details of his proposed method of maintaining the correct cover for approval by the S.O. prior to starting work.

- (ii) The number of joints in longitudinal steel bars shall be kept to a minimum. Joints in reinforcement shall be such that the full strength of each bar is effective across the joint. The reinforcement shall be adequately fixed to avoid displacement and to maintain the minimum specified cover during concreting.
 - (iii) The spacing of the steel reinforcement shall not be so close as to completely retain the concrete or to sufficiently impede the free flow of concrete to prevent the complete embedment of the reinforcement. The minimum clear spacing between the longitudinal steel bars shall be 100mm.
 - (iv) The Contractor shall ensure that reinforcement cages are extended at least 1 metre below the level of any temporary casing so as to prevent movement of the cage caused by the drag as the casing is extracted.
- (q) Concrete Mix
- (i) The Contractor shall take all precautions in the design of the mix and placing of the concrete to ensure that no arching of the concrete in a temporary casing will occur.
 - (ii) The concrete mix shall be designed to be self-compacting. Where placing the concrete using a tremie pipe is approved by the S.O., the minimum slump of the concrete shall be 150mm and the mix shall flow easily within the tremie pipe and be designed to produce a dense impervious concrete.
 - (iii) The Contractor shall design the concrete mix to achieve a slump measured at the time of discharge into the pile bore in accordance with the standards shown in table below.

Standards For Concrete Slump

Piling mix workability	Slump		Typical conditions of use
	Minimum, mm	Range, mm	
A	75	75-150	Placed into water-free unlined or permanently lined bore of 600mm diameter or over, or where casting level lies below the temporary casing; reinforcement widely spaced, leaving ample room for free movement of concrete between bars.
B	100	100-200	Where reinforcement is not spaced widely; where cut-off level of concrete is within temporary casing; where pile bore is water-free and the diameter is less than 600mm.
C	150	150 or more	Where concrete is to be placed by tremie under water or bentonite slurry or by pumping.

(r) Placing Concrete

- (i) The pile shall be concreted in one continuous operation without such interruption as would allow the previously placed batch of concrete to have hardened immediately after the bore excavation has been completed and inspected.
- (ii) The concrete shall be placed in the boring as soon as possible and in no case more than 30 minutes after mixing. The method of placing shall be such as to ensure that a continuous monolithic concrete shaft of the full cross-section is formed and that the concrete in its final position shall be dense and homogeneous.
- (iii) The Contractor shall ensure that no concrete impinges on either the reinforcement or on the sides of the pile bore so as to cause segregation of the concrete and inclusion of the dislodged soil debris.
- (iv) Adequate precautions shall be taken by the use of hopper or by other approved means to ensure that the concrete is not contaminated at the head of the shaft. The Contractor shall ensure that no soil, liquid or other foreign matter which would adversely affect the performance the pile shall be permitted to contaminate the concrete.
- (v) Internal vibrators shall not be used to compact concrete.
- (vi) The Contractor shall take measures to ensure that the structural strength of the concrete placed in all pile is not impaired through grout loss, segregation or bleeding when placing concrete in dry borings. Concrete shall be directed vertically into the centre of each vertical pile so that grout is not lost from the initial discharges.
- (vii) Before placing concrete under water or bentonite slurry, the Contractor shall ensure that there is no accumulation of silt or other material at the base of the boring, and the Contractor shall ensure that heavily contaminated bentonite suspension, which could impair the free flow of concrete from the tremie pile, has not accumulated in the bottom of the pile bore.
- (viii) If in the opinion of the S.O. it is not practical to keep pile bores free of water, the Contractor will be allowed to place concrete under water or bentonite slurry. Concrete to be placed under water or bentonite slurry shall be placed by tremie and shall not be discharged freely into the water or bentonite slurry.
- (ix) A sample of the bentonite suspension shall be taken from the base of the boring using an approved sampling device. If the specific gravity of the suspension exceeds 1.06 the placing of concrete shall not proceed. In this event the Contractor shall modify or replace the bentonite as approved to meet the Specification requirements.
- (x) The concrete shall be a rich coherent mix of high workability and cement content in accordance with the concrete general specification.
- (xi) The hopper and pipe of the tremie shall be clean and watertight throughout.

- (xii) The pipe shall extend to the base of the bore and a sliding plug or barrier shall be placed in the pipe to prevent direct contact between the first charge of concrete in the tremie and the water or bentonite slurry. The pipe shall at all times penetrate the concrete which has previously been placed and shall not be withdrawn from the concrete until completion of concreting. A sufficient quantity of concrete shall be maintained within the pipe to ensure that the pressure from it exceeds that from the water or bentonite slurry. The internal diameter of the pipe of the tremie shall be not less than 150mm for concrete made with 20mm aggregate. It shall be so designed that external projections are minimized, allowing the tremie to pass within reinforcing cages without causing damage. The internal face of the pipe of the tremie shall be free from projections. The details of the equipment and procedure for placing the concrete by tremie shall be submitted to the S.O. for approval.
- (s) Extraction of Casings
 - (i) Temporary casings shall be extracted while the concrete within them remains sufficiently workable to ensure that the concrete is not lifted. During extraction, the motion of the casing shall be maintained in an axial direction relative to the pile.
 - (ii) When the casing is being extracted, the Contractor shall maintain a sufficient quantity of concrete within it to ensure that pressure from external water, bentonite slurry or soil is exceeded by the concrete pressure and that the pile is neither reduced in section nor contaminated.
 - (iii) The concrete level within a temporary casing shall be topped up where necessary during the course of casing extraction in such a way that the base of the casing is always below the concrete surface until the casting of the pile has been completed.
 - (iv) The Contractor shall take adequate precautions in all cases where excess heads of water or bentonite slurry could occur as the casing is withdrawn because of the displacement of water or slurry by the concrete as it flows into its final position against the walls of the pile bore. Where two or more discontinuous lengths of casing (double casing) are used in the construction the proposed method of working shall be approved by the S.O..
- (t) Site Surface
 - (i) The Contractor shall allow for any surfacing, levelling, excavation and backfilling of the ground which he may require to carry out in the piling works
- (u) Pile Heads
 - (i) For piles cast in dry bores using temporary casing and without the use of a permanent lining, pile heads shall be cast to a level above the specified cut-off so that, after trimming, a sound concrete connection with the pile can be made. The casting level shall be within the tolerance above the cut-off level shown in Table G3.13-1, but shall not be above the commencing surface level.

- (ii) No pile shall be cast with its head below standing water level unless the Contractor has taken approved measures to prevent inflow of water causing segregation of the concrete as temporary casing is extracted; and, where applicable, the standing water level for each pile shall be agreed with the S.O., and this level is to be treated as the cut-off level for the purpose of calculating tolerance.

TABLE G3.13-1: Casting Tolerance Above Cut-Off For Piles Cast In Dry Bores Using Temporary Casing and Without The Use of A Permanent Lining

Cut-off distance below commencing surface, H, m	Casting tolerance above cut-off level, m
0.15 - 10.00	$0.3 + H/10$

If H is greater than C, then this tolerance is no longer applicable and the tolerances in Table G3.13-2 will apply.

- (iii) For piles cast in dry bores within permanent lining tubes or permanent casings, or where their cut-off levels are in stable ground below the base of any casing used, pile heads shall be cast to a level above the specified cut-off so that, after trimming, a sound concrete connection with the pile can be made. The casting level shall be within the tolerance above the cut-off level shown in the Table G3.13-2 but shall not be above the commencing surface level.

TABLE G3.13-2: Casting Tolerance Above Cut-Off Level For Piles Cast In Dry Bores Within Permanent Lining Tubes Or Permanent Casings, Or Where Their Cut-Off Levels Are In Stable Ground Below The Base Of Any Casing Used

Cut-off distance below commencing surface, H, m	Casting tolerance above cut-off level, m
0.15 - 10.00	$0.3 + H/12 + C/8$ where C = length of temporary casing below the commencing surface*

- (iv) For piles cast under water or bentonite slurry, the pile heads shall be cast to a level above the specified cut-off so that, after trimming to remove all debris and contaminated concrete, a sound concrete connection with the pile can be made. The casting level shall be within the tolerance above the cut-off level shown in table, but shall not be above the commencing surface level. Cut-off levels may be specified below the standing groundwater level, and where this condition applies the borehole slurry level shall not be reduced below the standing groundwater level until the concrete has set.

TABLE G3.13-3: Casting Tolerance Above Cut-Off for Piles Cast Under Water or Bentonite Slurry +

Cut-off distance below commencing surface, H, m	Casting tolerance above cut off level, m
0.15 - 10.00	$1.0 + H/12 + C/8$ where C = length of temporary casing below the commencing surface

In cases where a pile is cast so that the cut-off is within a permanent lining tube, the appropriate tolerance is given by deletion of the casing term C/8 in the table.

- (v) Where the cut-off level of piles lies at depths greater than 10m below the commencing surface, then the Contractor shall submit tolerances using that specified in Tables G3.13-1 to Table G3.13-3 as a guide for the S.O.'s approval before the commencement of the piling. In this Clause commencing surface is taken to mean the surface level at the pile position at the commencement of the boring operation.
- (vi) The pile heads will be cut off to the final levels.
- (vii) The cost of cutting or making good due to pile heads having been cast outside this tolerance or from defective concrete below the trimmed levels shall be borne by the Contractor.
- (viii) The length of the longitudinal reinforcement which shall be left projecting above the trimmed level of the pile shall be submitted to the S.O. for approval.
- (ix) After piling, the as-built positions of the piles shall be surveyed for eccentricities.
- (x) When the as-built eccentricities exceed 75mm in any direction, remedial actions shall be proposed by the Contractor's PE for S.O.'s approval before carrying out these actions.
- (xi) The cost of all the above shall be deemed to be included in the Tender Price and no claim whatsoever would be entertained by the Board.
- (v) Water Levels
 - (i) During extraction of temporary casings, where circumstances are such that newly placed unset concrete is brought into contact with external groundwater, the Contractor shall take necessary precautions to ensure that the internal concrete pressure at all levels within the pile exceeds the external groundwater pressure.
- (w) Backfilling Empty Borings
 - (i) Where the trimmed level of any pile is more than 600mm below the ground level, the surplus excavation shall be backfilled not more than 24 hours after the concreting is completed.

- (x) Disposal of Excavated Material
 - (i) The Contractor shall dispose off surplus earth resulting from excavation by removing it from the Site.
- (y) Records
 - (i) A complete record shall be maintained on Site of the nature of the strata through which each pile is bored together with the level of each change of stratum and the level of the base of the excavation. Such record and samples of strata drilled shall be submitted to the S.O. on completion of each pile.
- (z) Maximum Permissible Working Loads
 - (i) The Contractor shall install working piles to such depths of penetration as are necessary for each pile to develop a load carrying capacity of not less than the full test load (working load test) before failure occurs.
 - (ii) The maximum permissible working load of any pile shall not be more than the designed allowable strength (based on permissible working stresses) of the section used. This maximum shall be reduced as necessary depending on the nature of the site, the equipment used for installing the piles, the results of load tests, the likelihood of heaving and any other factor which may adversely affect the load carrying capacity of the pile.

G3.13.11 Steel Sheet Piling

- (a) Materials
 - (i) Steel Sheet Piling
 - 1) Steel for sheet piles forming part of the Permanent Works shall be grade S355GP in accordance with BS EN 10248.
 - (ii) Tie Rods
 - 1) Tie rods shall be of high yield steel having the following properties:
 - Minimum ultimate tensile strength 550 N/mm²
 - Minimum yield stress 410 N/mm²
 - Tie rods shall be supplied by a manufacturer approved by the S.O..
 - (iii) Welding Consumables
 - 1) All welding consumables (electrodes, wire, filler rods, flux, shielding gas and the like) shall comply with the requirements of BS EN 1011-2.
 - 2) In addition, weld electrodes for metal-arc welding shall conform with BS EN 499 and to the requirements of the appropriate weld procedure.

(iv) Steel Sheet Piling Sections

- 1) Unless otherwise shown on the Drawings, steel sheet pile sections shall be Frodingham or other section approved by the S.O.. Where pile sections other than those shown on the Drawings are approved for use, the Contractor shall be responsible for producing revised layout drawings for the approval of the S.O.. Piles shall be straight and neither twisted about the longitudinal axis nor having the transverse section distorted to an extent that prevents piles interlocking.
- 2) Steel sheet piles shall be capable of being interlocked, one with another, to form a continuous wall or diaphragm. Where more than one shape of section is provided, transition pieces shall be provided to ensure continuity. The change from one section shape to another shall where possible take place at a wall corner. Special corner plates shall be provided at all abrupt changes of direction.
- 3) Steel sheet piles shall be rolled, shaped and cut to length under factory conditions. Maximum rolling margins shall be 4 % above and 2.5 % below the calculated masses, and 75 mm over and 50 mm under the required lengths.

(v) Caulking

- 1) Caulking material shall, according to the gap in the pile clutch, be 5 mm diameter 'Denso twist', 2.5 mm 'Denso cord', 'Denso tape' (all manufactured by Winn & Coales Ltd of Denso House, Chapel Road, London SE27 OTR) or other approved material.

(vi) Tie-rod Protection

- 1) Materials for protecting tie rods shall be manufactured by Winn & Coale or other approved materials.

(b) Workmanship

(i) Storage and Handling

- 1) Before taking delivery of steel sheet piles, the Contractor shall submit to the S.O. for approval detailed proposals for storing, handling and transporting the piles. If required by the S.O., he shall either modify the details or submit new proposals.
- 2) Steel sheet piles which are of different types, cross-section, hand or length shall be stored in separate stacks. Piles shall be supported clear of the ground on suitable bearers and bearers shall be placed where necessary between piles in the stack. Piles may be stacked either on top of each other, with the longitudinal axes parallel so that the piles rest together, or with a number of piles side by side in layers, with the longitudinal axes of the piles in alternate layers placed orthogonally. The number of layers of piles resting on a pile shall be suitably limited. It shall be possible to lift piles freely from the stack without gripping the piles remaining in the stack.

- 3) Piles shall be lifted using only lifting holes and standard shackles or slinging attachments provided or recommended by the manufacturer of the piles. Secondary lifting point or equivalent prevention method shall be provided in the event the main lifting holes or sling fail.
- 4) Piles shall be transported on suitable vehicles or trailers and shall be adequately supported and properly secured so that the piles are not distorted or damaged in transit.

(ii) Driving Alignment

- 1) The Contractor shall drive piles to the lines, levels and dimensions shown on the Drawings. Finished piles shall not depart from their correct positions by more than the following amounts where:
 - 'The line of the piles' at any level means the line through the points of intersection of the axes of those piles and the horizontal plane at that level;
 - 'The plane of the piles' means the plane containing the axes of those piles; and
 - 'True' means in accordance with the Drawings.
- 2) The line of the piles shall not vary from the true line by more than 50 mm in plan.
- 3) The inclination of the plane of the piles shall not vary from the true inclination by more than 1 in 100. The inclination of any pile axis in that plane shall not vary from the line of greatest slope in that plane by more than 1 in 200.
- 4) The finished level of the top of any pile shall not vary from the true level by more than 20 mm.
- 5) The finished level of the bottom of any pile shall not vary from the true level by more than 1 % of the specified length of the pile. Piles shall be prevented from being drawn below the true level during the driving of neighbouring piles.
- 6) Adjacent piles may be rotated relative to each other about their clutches by not more than $41/2$ degrees in the case of Frodingham sections, and not more than the amount recommended by their manufacturers in the case of other types of pile. The rotation shall be only for the purposes of accommodating horizontal curves and minor changes in alignment or, when piling with Frodingham section piles, for increasing or decreasing the number of piles in a length of sheet pile wall as necessary to suit the required arrangements.

(iii) Extended Piles

- 1) Piles shall not be extended without the approval of the S.O., the Contractor may cut off the top of a pile to conform to the required level.

(iv) Engagement of Pile Clutches

- 1) Unless otherwise approved by the S.O., clutches of piles shall be engaged properly for the full length of the piles. Piles which fail to comply with this provision shall be extracted but, provided that they subsequently comply with the specified requirements for straightness, they may be re-driven.

(v) Construction Plant and Methods

- 1) The Contractor shall be responsible for the choice of the Contractor's equipment and Temporary Works for sheet piling. Pile-driving and extracting equipment may be of either percussive or vibratory type, provided that the Contractor demonstrates to the satisfaction of the S.O., that the methods and equipment proposed, do not result in ground resonance which might adversely affect neighbouring existing structures or work under construction. The equipment shall be sufficiently robust, heavy and powerful to drive and extract the piles as specified without damaging them, and the size and type of equipment shall be changed as necessary when driving a pile. Suitable driving caps, helmets, dollies and anvil blocks shall be provided which shall be the correct size to suit the section of the pile to be driven and to avoid distorting or damaging the pile. Should distortion or damage occur, the defective portion of pile shall be cut off. Pile grips shall grip the pile tightly. The direction of the driving forces produced by pile driving and extracting equipment shall be parallel to the longitudinal axis of the pile. Except where a suitable cradle is provided for mounting the pile driving equipment on the side of a pile, pile driving and extracting equipment shall normally be placed centrally upon a pile or piles. Suitable leg-guides, piling frames or guide frames of adequate rigidity and stability to hold piles firmly in place during driving shall be provided. The vertical spacing between guides to accommodate the pile shall be as small as possible and shall, in any case, not exceed by more than 10 mm the overall thickness of the pile.
- 2) At all times during driving, piles shall be firmly held within the guides by suitable packing pieces, which shall not be wedges. Frodingham-section piles shall be provided with packing pieces for each pile. Suitable arrangements shall be made to accommodate bends, curves, relative rotation in adjacent piles, and special piles, as required.

- 3) Unless otherwise approved by the S.O., standard piles shall be driven in pairs and guided in panels comprising a limited number (not exceeding 20) of pairs of piles. The pair of piles at each end of a panel shall be partially driven to an adequate depth so that additional guidance during the driving is provided for the intermediate pairs of piles in the panel, which shall be pitched and held in place ready for driving. The intermediate pairs of piles shall be driven subsequently in turn in such a manner that the penetration of the bottom of an intermediate pair beyond that of an adjacent intermediate pair is suitably limited to ensure that adequate support and guidance is at all times provided by the adjacent pair. The panel shall be driven in stages, the number and dimensions of the stages being related to the lengths and sections of the piles driven, and in such a way that the guides remain in position as long as possible, being removed in turn at the end of each stage as necessary to allow the pile-driving equipment to drive the subsequent stage. At the end of a panel, the partially driven pairs may remain in the partially driven position to provide guidance during subsequent driving of the adjacent panel and may be used for the support of the guides for the adjacent panel. Deviation from verticality shall not be corrected by means of props or tensioned ties other than the approved supports and guides.
- 4) Pile driving shall not be assisted by jets of water or air directed into the ground except with the approval of the S.O..
- 5) Until the driving of a pile is completed, the Contractor shall not excavate below the existing ground surface adjacent to the pile unless such excavation is either approved or ordered by the S.O..
- 6) If there is any defect in the horizontal or vertical alignment of the piling, or if a break through any cause whatsoever occurs or is suspected to have occurred in the continuity of the piling, the Contractor shall withdraw such pile or piles as may be necessary to remedy the defect or suspected defect. If any distortion, creep or fanning occurs in the line of the piles, the Contractor shall take steps to rectify the defect and shall where necessary supply and drive suitable tapered piles to the satisfaction of the S.O..

(vi) Piles in Trenches

- 1) The Contractor may, with the approval of the S.O., excavate a trench in which the steel sheet piling is subsequently pitched and driven. If such a trench is allowed to be excavated, the void between the piles and undisturbed ground shall be backfilled with concrete grade C25/20 immediately after driving of the piles is complete.

(vii) Cutting Piles

- 1) Steel sheet piles shall be cut either by using suitable flame cutting equipment or by sawing, mechanically or by hand, without flames or sparks, as directed or approved by the S.O.. Off-cuts shall be either disposed of off the Site or, if required and suitable for use in the Permanent Works, stored on Site.

(viii) Holes in Piles

- 1) Steel sheet piles in the Permanent Works shall be free from holes other than holes specified on the Drawings or small holes less than 40 mm diameter and not more than 200 mm from the top of the piles to assist in pitching the piles prior to driving. No other holes will be permitted without the prior approval of the S.O.. Where such approval is given, the holes shall subsequently be patched with steel plate of equal thickness to the material removed. As directed by the S.O., either a single patch welded continuously on both sides or two patches applied one on each side of the pile and continuously welded shall be used.
- 2) Holes may be drilled or flame cut, but the size of the hole as formed measured in any direction shall not exceed by more than 20% the diameter of the required circular hole. All holes for bolts, tie rods and drains shall be drilled or flame cut after the sheet piling has been driven.

(ix) Welding of Piles

- 1) All welding of piles shall be in accordance with the requirements of the relevant standard. Welded joints in adjacent spliced piles shall be staggered by not less than 0.6 m. Details of the proposed weld procedures shall be submitted to the S.O. for approval. All slag shall be removed and any sharp projections shall be ground smooth.
- 2) All welders shall pass qualification tests relevant to the weld procedures in use in accordance with BS EN 287-1. Welders shall produce satisfactory evidence of having been engaged in welding for at least 9 months in the preceding 12 month period. If the work of any welders employed on the Contract is unsatisfactory, the Contractor shall carry out such further welder qualification tests as are necessary to demonstrate that the welders are proficient.
- 3) Welds shall be subjected to non-destructive testing by processes which may include but shall not necessarily be limited to radiographic, ultrasonic, magnetic particle, or dye penetrant methods, depending on the type of weld and its position in the structure.
- 4) If any work shows defects or fails to comply with the requirements of the Specification for any reason, it shall be repaired or rejected, even though it may have been carried out by qualified welders using approved procedures.

(x) Watertightness of Piling

- 1) Steel sheet piling against which concrete is to be cast shall be clean and watertight. All such steel sheet piling shall be free from loose rust, loose millscale, oil, grease and earth or any other substance which, in the opinion of the S.O. is likely to prevent bonding between the concrete and the piles. All such piling shall be continuously caulked with an approved caulking material on the excavation side of the sheeting. The Contractor shall use a material which suits the gap in the clutch of the pile and shall drive it into the gap, using a hammer and blade or caulking tool. Pile clutches shall be prepared for caulking in accordance with the recommendations of the manufacturer of the caulking material.

(xi) Protection of Piling

- 1) The surface of steel piles and permanent walings which will not, on completion of the Works, be buried in the ground or in concrete, shall, after being exposed, be blast cleaned and painted with black bituminous paint applied in two coats: the first applied within 2 hours of blast cleaning and the second within 24 hours of application of the first coat.

(xii) Protection of Tie Rods

- 1) Tie rods shall be thoroughly wire brushed and then primed by spreading 'Denso paste' or similar over the entire surface including the surface of threads, if any. Areas between nuts and rods shall be bevelled with 'Densylmastic' or similar, to present a rounded profile. The entire surface of the assembled tie rod shall then be spirally wound with 'Denso tape' or similar allowing a 55 % overlap. The tape shall be pressed into intimate contact with the underlying paste. The entire taped surface shall then be over-wrapped with 'Denso PVC self-adhesive' tape or similar, again allowing a 55 % overlap.
- 2) After wrapping, tie rods shall be sleeved with 200 mm diameter uPVC pipes.

(xiii) Tie Rod Anchorages

- 1) The forces in the tie rods will be transmitted by means of nuts and washers on the threaded ends of the tie rods bearing onto walings at the pile end and onto steel anchor plates at the anchor block. Threads on tie rods shall be rolled on tie rods not exceeding 75 mm diameter and shall be cut on tie rods of larger diameter.
- 2) Anchor blocks for tie rods shall be made in reinforced concrete, grade C30/40. The bottom and thrust surfaces of the concrete shall be cast against polythene sheet placed in direct contact with the excavated face. Polythene sheet shall be wrapped and sealed around the remaining faces of the block, including the tie rod anchorages.

(xiv) Concrete Capping Beams

- 1) Concrete capping beams shall be formed of reinforced concrete, grade C32/40, cast in situ. The concrete thickness on each side of the sheet piles may vary due to the permitted tolerance in pile or formwork position but the total thickness of the capping beam shall not vary.

- 2) If the Contractor drives piles to tolerances exceeding those specified, the Contractor shall submit to the S.O. for approval his proposals for constructing the capping beam to accommodate the piles driven outside the permitted tolerance. If the Contractor's proposals include constructing the capping beam to an increased thickness, the increase shall be maintained along the full length of the capping beam at no extra cost to the Board.

(xv) Piling Records

- 1) The Contractor shall maintain a full record of all steel piling work. This record shall include the following:
 - Identification number for each pile;
 - Pile section;
 - Pile size;
 - Description of special piles;
 - Original length;
 - Supplementary length;
 - Final length;
 - Levels of any obstructions met in piling;
 - Levels at which any substantial change in rate of penetration occurred;
 - Number of blows, or time, to penetrate 100 mm at one metre intervals in the final 4 metres of driving;
 - Type, size and rated energy of pile-driving plant; and
 - Stage driven by each type of plant.

A copy of the records of each day's piling shall be submitted to the S.O. before the next day's piling begins or within 24 hours of the end of the day's piling whichever shall be sooner.

END OF SECTION